Formalization of Formal Group Laws

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1 Formal Group Laws

1.1 Definition and Basic Properties of One-Dimensional Formal Group Laws

Definition 1.1.1 ([6, Definition 1.1])**.** Let R be a commutative ring. A one-dimensional formal group law is a formal power series $F(X,Y) \in R[\![X,Y]\!]$ with the following properties:

1. $F(X,Y)$ is of the form
$$F(X,Y) = X + Y + \sum_{i,j\geq 1} c_{ij} X^i Y^j \tag{1.1.1.1}$$

2. $F(X,Y)$ satisfies the associativity condition, namely
$$F(F(X,Y),Z) = F(X,F(Y,Z)) \tag{1.1.1.2}$$

Furthermore, if $F(X,Y)$ satisfies $F(X,Y) = F(Y,X)$, then it is called a commutative formal group law.

Remark 1.1.2. Under this definition, one can prove that $F(X,0) = X$ and $F(0,Y) = Y$, which are used as part of the definition in other references. Therefore, given a formal group law $F(X,Y) \in R[\![X,Y]\!]$, it can be written as $F(X,Y) = X + Y * G_1(X,Y)$ and $F(X,Y) = Y + X * G_2(X,Y)$, for some $G_1(X,Y), G_2(X,Y) \in R[\![X,Y]\!]$.

Proof of equivalence of the definition. Let $f(X) = F(X,0)$. Then we have
$$F(F(X,0),0) = F(X,F(0,0)) = F(X,0),$$
which means that $f \circ f = f$. Composing both sides with f^{-1} gives $f = \mathrm{id}$. This is the desired result for $F(X,0) = X$. Also, $F(0,X) = X$ follows from a similar proof. □

Example 1.1.3. There are two basic commutative formal group laws:

$$\mathbf{G}_a(X,Y) = X + Y \quad \text{(the additive formal group law.)} \tag{1.1.3.1}$$
$$\mathbf{G}_m(X,Y) = X + Y + XY \quad \text{(the multiplicative formal group law.)} \tag{1.1.3.2}$$

Lemma 1.1.4. *Let $F(X,Y)$ be a formal group law over a ring R. Then there exists a power series $i(X) = -X + b_2X^2 + \cdots \in R[\![X]\!]$ such that $F(X,i(X)) = 0$ and $F(i(X),X) = 0$.*

In particular, the left inverse and right inverse are the same and the inverse is unique.

Proof. Let $i(X) = \sum_{j=1}^{\infty} b_j X^j$, and choose b_j inductively so that $F(X,i(X)) = 0$.

First, choose $b_1 = -1$. Then $F(X,i(X)) = 0 \mod \deg 2$. Assume that $b_1,\ldots,b_n$ are already defined and $F(X,i(X)) = 0 \mod \deg(n+1)$. Then choose b_{n+1} to be minus the $(n+1)$-th coefficient of $F(X,i_n(X))$, where $i_n(X) = \sum_{j=1}^{n} b_j X^j$. Therefore, $F(X,i_{n+1}(X)) = 0 \mod \deg(n+2)$, where $i_{n+1}(X) = i_n(X) + b_{n+1}X^{n+1}$. Repeating this process, we obtain

the desired series $i(X)$ with $F(X, i(X)) = 0$. We can use a similar process to define $i'(X)$ and $F(i'(X), X) = 0$.

Since $F(X, Y)$ is a formal group law, we have

$$i'(X) = F(i'(X), 0) = F(i'(X), F(X, i(X))) = F(F(i'(X), X), i(X)) = F(0, i(X)) = i(X)$$

Therefore, the left inverse and right inverse of X are the same. □

Combining the result above, a formal group law $F(X, Y)$ defines a group structure on multivariate power series in $R[\![X_I]\!]$ with zero constant coefficient, where I is any finite index set.

Definition 1.1.5 ([6, Definition 1.4])**.** Let $F(X, Y)$ and $G(X, Y)$ be two formal group laws over a ring R. A *homomorphism* $F(X, Y) \to G(X, Y)$ is a power series $\alpha(X) = a_1X + a_2X^2 + \cdots \in R[\![X]\!]$ such that

$$\alpha(F(X, Y)) = G(\alpha(X), \alpha(Y)) \tag{1.1.5.1}$$

An endomorphism is a homomorphism $F(X, Y) \to F(X, Y)$. The homomorphism α is an *isomorphism* if there exists a homomorphism $\beta(X) : G(X, Y) \to F(X, Y)$ such that $\alpha(\beta(X)) = \beta(\alpha(X)) = X$.

Lemma 1.1.6. *The homomorphism $\alpha(X) : F(X, Y) \to G(X, Y)$ is an isomorphism if and only if a_1 is a unit in the ring, where $\alpha(X) = a_1X + a_2X^2 + \cdots$.*

Proof. Given that $\alpha(X)$ is a power series with a_1 a unit, there exists a power series $\beta(X)$ such that $\alpha(\beta(X)) = \beta(\alpha(X)) = X$. We need to prove that this $\beta(X)$ is a formal group homomorphism from $G(X, Y)$ to $F(X, Y)$, namely $\beta(G(X, Y)) = F(\beta(X), \beta(Y))$.

Since $\alpha(X)$ is a homomorphism from $F(X, Y)$ to $G(X, Y)$,

$$\alpha(F(\beta(X), \beta(Y))) = G(\alpha(\beta(X)), \alpha(\beta(Y))) = G(X, Y).$$

The desired equality follows by pre-composing the above equation with $\beta(X)$. □

One of the most important endomorphisms of a formal group is the n-series.

Definition 1.1.7 (n-series [6, Definition 1.4.5])**.** For any natural number n, we define the n-series $[n](X) \in R[\![X]\!]$ as follows:

1. $[0](X) = 0$.
2. $[n + 1](X) = F([n](X), X)$.

Moreover, $[-n](X) = i([n](X))$, where $i(X)$ is the power series in Lemma 1.1.4.

Proposition 1.1.8. *For any natural number n, n-series $[n](X)$ is a formal group homomorphism from $F(X, Y)$ to $F(X, Y)$, i.e. $[n](F(X, Y)) = F([n](X), [n](Y))$.*

Proof. We use induction on n to prove it. For the base cases $n = 0, 1$, the result follows immediately. Assume that the result holds for some natural number k. Now consider the case $k + 1$.

$$\begin{aligned}
[k + 1]F(X, Y) &= F([k]F(X, Y), F(X, Y)) \\
&= F(F([k](X), [k]Y), F(Y, X)) \\
&= F([k](X), F([k](Y), F(Y, X)))
\end{aligned}$$

Next, we focus on the term $F([k](Y), F(Y, X))$.

$$F([k](Y), F(Y, X)) = F(F([k](Y), Y), X) = F([k+1](Y), X) = F(X, [k+1](Y))$$

Hence, we have

$$\begin{aligned}[k+1]F(X, Y) &= F([k](X), F(X, [k+1](Y))) \\ &= F(F([k](X), X), [k+1](Y)) \\ &= F([k+1](X), [k+1](Y))\end{aligned}$$

□

Remark 1.1.9. By induction, we deduce that $[n](X) = nX + O(X^2)$. As a result, for a ring R of characteristic p, we have $[p](X) = aX^k + O(X^{k+1})$ for some $k > 1$.

Proposition 1.1.10. *Assume R is of characteristic p. Then $[p](X) = 0$ or $[p](X) = aX^{p^n} + O(X^{p^n+1})$.*

Proof. The proof can be found in Corollary 1.5.5. □

Definition 1.1.11 ([6, Definition 1.5])**.** Let $F(X, Y)$ be a formal group law over a ring R, and let $\phi : R \to R'$ be a ring homomorphism. Suppose $F(X, Y) = X + Y + \sum_{i,j\geq 1} c_{ij}X^iY^j$. By applying ϕ to the coefficients of $F(X, Y)$, we obtain a formal group law

$$\phi_* F(X, Y) = X + Y + \sum \phi(c_{ij})X^iY^j \tag{1.1.11.1}$$

Remark 1.1.12. There is a functor from the category of commutative rings to **Type**, which maps a commutative ring R to all formal group laws over R. Given a ring homomorphism $\phi : R \to S$, the definition above provides a map from formal group laws over R to formal group laws over S.

1.2 Functional Equation Lemma

In this section, we discuss the functional equation lemma for formal group laws. This lemma can be used to construct some non-isomorphic formal group laws and a universal formal group law.

To begin, we need the following ingredients. Let R be a subring of K, $\sigma : K \to K$ be a ring homomorphism, I be an ideal of R, p be a prime number, q be a power of this prime number, and $s_1, s_2, \ldots$ be elements of K.

These ingredients are assumed to satisfy the following conditions:

1. $\sigma(R) \subset R$
2. $\forall a \in R, \sigma(a) \equiv a^q (\bmod I)$
3. $p \in I$
4. $\forall i,\ s_i I \subset R$
5. $\forall b \in K, r \in \mathbb{N},\ I^r b \subset I \Rightarrow I^r \sigma(b) \subset I$.

Definition 1.2.1 ([6, Definition 2.1.8])**.** Let $g(X) = \sum_{i=1}^{\infty} b_i X^i$ be a power series in one variable with coefficients in R, then define the **recursion function** to be the unique solution of the following functional equation,

$$f_g(X) = g(X) + \sum_{i=1}^{\infty} s_i \sigma_*^i f_g(X^{q^i}) \tag{1.2.1.1}$$

where $\sigma_*^i f_g(X)$ is the power series obtained from $f_g(X)$ by applying the endomorphism σ^i to the coefficients.

Remark 1.2.2. If we denote $f_g(X) = \sum_{i=1}^{\infty} a_i X^i$ and $n = q^r m$, q does not divide m. By comparing coefficients on both sides, there is a recursive formula for the coefficients of $f_g(X)$.

$$a_n = b_n + s_1 \sigma(a_{n/q}) + \cdots + s_r \sigma^r(a_{n/q^r}) = b_n + \sum_{i=1}^{r} s_i \sigma^i(a_{n/q^i}) \tag{1.2.2.1}$$

In particular, if q does not divide n, then $a_n = b_n$.

The functional equation lemma concerns the integrality of the coefficients of $f_g(X)$ and related power series.

Theorem 1.2.3 (Functional Equation Lemma [6, Chap. 1, Theorem 2.2])**.** *Let $g(X)$, $h(X)$ be two power series in one variable over R, suppose that the coefficient of X in $g(X)$ is invertible in R, then*

1. *the power series $F_g(X, Y) = f_g^{-1}(f_g(X) + f_g(Y))$ has its coefficients in R.*
2. *the power series $f_g^{-1}(f_h(X))$ has its coefficients in R.*
3. *there is a power series $\hat{h}(X)$ over R such that $f_g(h(X)) = f_{\hat{h}}(X)$.*
4. *let $\alpha(X) \in R[\![X]\!]$, $\beta(X) \in K[\![X]\!]$, and let r be a positive natural number, then we have*

$$\begin{aligned} &\alpha(X) \equiv \beta(X) (\operatorname{mod} I^r R[\![X]\!]) \\ \Leftrightarrow\ &f_g(\alpha(X)) \equiv f_g(\beta(X)) (\operatorname{mod} I^r R[\![X]\!]) \end{aligned}$$

Remark 1.2.4. According to the functional equation lemma, $F_g(X, Y)$ is a commutative formal group law over the ring R.

Before discussing the proof the functional equation integrality lemma, we first present several technical lemmas.

Lemma 1.2.5 ([6, Chap. 1, Lemma 2.4.1])**.** *Let $f_g(X) = \sum_{i=1}^{\infty} a_n X^n$, and let $n = q^r m$ where m is not divisible by q, then $a_n I^r \subset R$.*

Proof. By the assumption 5 and induction, we have that for any given natural number j, r and $b \in K$,

$$I^r b \subset I \Rightarrow I^r \sigma^j(b) \subset I. \tag{1.2.5.1}$$

We do induction with respect to r (multiplicity of q in n).

Assume that the result holds for all numbers less than $r+1$; now consider those n such that the multiplicity of q in n is $r+1$. According to our recursive formula, the coefficient a_n can be expressed as

$$a_n = b_n + \sum_{i=1}^{r+1} s_i \sigma^i(a_{n/q^i}).$$

Combining the result 1.2.5.1 and the assumption 4, it is enough to prove

$$\forall i \in [1, r+1], a_{n/q^i} I^{r+1} \in I.$$

According to the induction hypothesis and the fact that the multiplicity of q in n/q^i is less than $r+1$, then $\forall i \in [1, r+1], a_{n/q^i} I^r \in R$. This implies that $a_n \in I$. □

Lemma 1.2.6. *Let $F(X_J), G(X_J) \in R[\![X_J]\!]$, where J is a finite index set, let r be the multiplicity of q in n, namely $n = q^r m$ with $q \nmid m$ and i be a positive natural number, assume that $F(X_J) \equiv G(X_J) (\bmod I^i)$, then we have*

$$F(X_J)^n \equiv G(X_J)^n (\bmod I^{i+r}). \tag{1.2.6.1}$$

Proof. First we prove that $F(X_J)^{q^r} \equiv G(X_J)^{q^r} (\bmod I^{i+r})$, and then our goal is followed by taking some power at both sides.

The case for $r = 0$ is trivial. We do induction on r. Assume that the result holds for r, namely $F(X_J)^{q^r} \equiv G(X_J)^{q^r} (\bmod I^{i+r})$. Therefore, $F(X_J)^{q^r} = G(X_J)^{q^r} + H(X_J)$, where all coefficients of $H(X_J)$ are in I^{i+r}. And we have the following equality

$$\begin{aligned} F(X_J)^{q^{r+1}} = (F(X_J)^{q^r})^q &= (G(X_J)^{q^r} + H(X_J))^q \\ &= (G(X_J)^{q^r})^q + H(X_J)^q + qG(X_J)^{q^r} H(X_J)\alpha(X_J) \end{aligned}$$

for some $\alpha(X_J) \in R[\![X_J]\!]$. And the last equality follows from q is a prime power. Our result is given by the following two congruences $H(X_J)^q \equiv 0 (\bmod I^{i+r+1})$ and $qH(X_J) \equiv 0 (\bmod I^{i+r+1})$. □

Lemma 1.2.7 (Generalization of [6, Chap. 1 Lemma 2.4.2])**.** *For all $G(X_J) \in R[\![X_J]\!]$, where J is a finite index set, and let r be the multiplicity of q in n, namely $n = q^r m$ with $q \nmid m$, then*

$$G(X_J)^{nq^l} \equiv (\sigma_*^l G(X_J^{q^l}))^n (\bmod I^{r+1}) \tag{1.2.7.1}$$

Proof. The case for $I = R$ is trivial. Now, we assume that $I \subsetneq R$. Then R/I has characteristic p, since $p \in I$. Given property 2, then we have

$$G(X_J)^{q^l} \equiv \sigma_*^l G(X_J^{q^l}) (\bmod I)$$

We get the goal congruence by above congruence and lemma 1.2.6. □

Proof of Functional Equation Lemma. For simplicity, we write $F(X,Y)$ and $f(X)$ for $F_g(X,Y)$ and $f_g(X)$ respectively. Let $F_i(X,Y)$ be the homogeneous component of $F(X,Y)$ of degree i. And we denote $g(X) = \sum_{i=1}^{\infty} b_i X^i$ and $f(X) = \sum_{n=1}^{\infty} a_n X^n$ throughout the proof of this theorem.

Proof of part 1: Since $f(X) \equiv b_1 X (\bmod \deg\ 2)$ and $f^{-1}(X) \equiv b_1^{-1} X (\bmod \deg\ 2)$. Hence $F(X,Y) = X + Y (\bmod \deg\ 2)$, i.e. $F_1(X,Y)$ has its coefficients in R. We now proceed by induction to prove that $F_n(X,Y) \in R[\![X,Y]\!]$.

Suppose that $F_1(X,Y), \ldots, F_{n-1}(X,Y) \in R[\![X,Y]\!]$. And we have

$$(F_1(X,Y) + \cdots F_{n-1}(X,Y))^r \equiv (F(X,Y))^r (\bmod \deg\ (n+1)).$$

By the definition of $F(X,Y) = f^{-1}(f(X) + f(Y))$, we have

$$\begin{aligned} f(F(X,Y)) &= f(X) + f(Y) \\ \sigma_*^i f(\sigma_*^i F(X,Y)) &= \sigma_*^i f(X) + \sigma_*^i f(Y). \end{aligned}$$

by the recursion formula, $f(X) = g(X) + \sum_{n=1}^{\infty} s_n \sigma_*^n f(X^{q^n})$, substituting $F(X,Y)$ into X, and writing $f(X) = \sum_{n=1}^{\infty} a_n X^n$, we obtain that

$$f(F(X,Y)) = g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(F(X,Y))^{q^i n} \tag{1.2.7.2}$$

Next by the second technical lemma, we have

$$s_i \sigma^i(a_n) F(X,Y)^{q^i n} \equiv s_i \sigma^i(a_n)(\sigma_*^i F(X^{q^i}, Y^{q^i}))^n (\bmod (R, \deg(n+1))) \tag{1.2.7.3}$$

then we have that under the sense of $(\bmod (R, \deg(n+1)))$

$$\begin{aligned} f(F(X,Y)) &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(\sigma_*^i F(X^{q^i}, Y^{q^i}))^n \\ &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(\sigma_*^i F(X^{q^i}, Y^{q^i})) \\ &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i (\sigma_*^i f(X^{q^i}) + \sigma_*^i f(Y^{q^i})) \\ &\equiv g(F(X,Y)) + f(X) + f(Y) - g(X) - g(Y) \end{aligned}$$

Since $g(X) \equiv b_1 X (\bmod \deg\ 2)$, then

$$g(F(X,Y)) \equiv b_1 F_n(X,Y) (\bmod R, \deg(n+1))$$

use the equation $f(F(X,Y)) = f(X) + f(Y)$, and the congruence above, we have that

$$b_1 F_n(X,Y) \equiv 0 (\bmod R, \deg(n+1))$$

and b_1 is invertible in R, then we get the result.

Proof of part 2: Write $G(X) = f_g^{-1}(f_h(X))$. Let $G_i(X)$ be homogeneous component of degree i in $G(X)$. We prove that for all i, G_i has its coefficient in R. Assume that this result holds for

all $i < n$, now consider the case for $i = n$, then we have under the sense of $(\bmod R, \deg(n+1))$.

$$\begin{aligned}
f_h(X) &= f(G(X)) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(G(X)^{q^i}) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i \sum_{i=1}^{\infty} a_n G(X)^{q^i n} \\
&\equiv g(G(X)) + \sum_{i=1}^{\infty} s_i \sum_{i=1}^{\infty} \sigma_*^i(a_n) \sigma_*^i G(X^{q^i})^n \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(G(X^{q^i})) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i (f_h(X^{q^i})) \\
&= g(G(X)) + f_h(X) - h(X)
\end{aligned}$$

Then we have that $g(G(X)) \equiv h(X)(\bmod R, \deg(n+1))$. And since

$$h(X) \equiv g(G(X)) \equiv b_1 G_n(X).(\bmod R, \deg(n+1)).$$

Since b_1 is invertible, then $G_n(X)$ has its coefficient in R.

Proof of part 3: Denote $\hat{f}(X) = f(h(X))$. By the functional equation 1.2.1.1, we only need to prove that $\hat{f}(X) - \sum_{i=1}^{\infty} s_i \sigma_*^i \hat{f}(X^{q^i})$ has all its coefficients in R. We have the following congruence:

$$\begin{aligned}
\hat{f}(X) - \sum_{i=1}^{\infty} s_i \sigma_*^i \hat{f}(X^{q^i}) &= f(h(X)) - \sum_{i=1}^{\infty} s_i \sigma_*^i f(\sigma_*^i h(X^{q^i})) \\
&= f(h(X)) - \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(\sigma_*^i h(X^{q^i}))^n \\
&\equiv f(h(X)) - \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(h(X)^{q^i n}) \\
&= f(h(X)) - \sum_{i=1}^{\infty} s_i \sigma_*^i f(h(X)^{q^i}) \\
&= g(h(X)) \equiv 0
\end{aligned}$$

All congruences above are understood modulo R. The last term has all coefficients in R.

Proof of part 4: For the forward implication: assume that $\alpha(X) \equiv \beta(X)(\bmod I^r R[\![X]\!])$, then $\beta(X) \in R[\![X]\!]$. According to the lemma 1.2.6, we have that $\alpha(X)^n \equiv \beta(X)^n \mod I^{r+t}$, where t is the multiplicity of q in n. Let $f(X) = \sum_{i=1}^{\infty} a_i X^i$, according to the lemma 1.2.5, we have that $a_n \alpha(X)^t \equiv a_n \beta(X)^t \mod I^r$, where t is multiplicity of q in n. Therefore, $f(\alpha(X)) \equiv f(\beta(X)) \mod I^r$.

For the reverse direction, assume that $f(\alpha(X)) \equiv f(\beta(X))(\bmod I^r R[\![X]\!])$.

First, we prove the following result,

$$\alpha(X) \equiv 0(\bmod I^r) \Rightarrow f^{-1}(\alpha(X)) \equiv 0(\bmod I^r) \tag{1.2.7.4}$$

Let $\gamma(X) = f^{-1}(\alpha(X))$, then $\alpha(X) = f(\gamma(X))$. Let $\gamma_n(X)$ be the truncation of degree $n+1$ of $\gamma(X)$, and we use induction to prove all $\gamma_n(X) \in I^r R[\![X]\!]$. The cases $k = 1, 2$ are trivial. Assume that the result holds for k, namely $\gamma_k(X) \equiv 0 (\operatorname{mod} I^r)$; now we consider $\gamma_{k+1}(X)$. According to the functional equation and definition of $\gamma(X)$, we have

$$\alpha(X) = g(\gamma(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(\gamma(X)^{q^i}) \tag{1.2.7.5}$$

We claim that:

$$\sum_{i=1}^{\infty} s_i \sigma_*^i f(\gamma(X)^{q^i}) \in I^r R[\![X]\!] (\operatorname{mod} \deg(k+2)). \tag{1.2.7.6}$$

And we consider the $k+1$-th coefficient at both equation 1.2.7.5, then we have $b_1 * c_{k+1} \in I^r$, where $\gamma(X) = \sum_{i=1}^{\infty} c_i X^i$ and $g(X) = \sum_{i=1}^{\infty} b_i X^i$. Since b_1 is a unit in R, then $c_{k+1} \in I^r$.

Proof of claim 1.2.7.6 above. According to the lemma 1.2.6 and hypothesis assumption, we have that $\gamma_k(X)^n \equiv 0 (\operatorname{mod} I^{r+t})$, where t is multiplicity of q in n. By the assumption 4, it is enough to prove that

$$\sigma_*^i f(\gamma(X)^{q^i}) \in I^{r+1} R[\![X]\!] (\operatorname{mod} \deg(k+2)) \tag{1.2.7.7}$$

which is equivalent to

$$\sigma^i(a_n) \gamma(X)^{q^i n} \in I^{r+1} R[\![X]\!] (\operatorname{mod} \deg(k+2)) \tag{1.2.7.8}$$

Since

$$\gamma(X)^{q^i n} \equiv \gamma_k(X)^{q^i n} (\mod \deg(k+2))$$

and lemma 1.2.5, we have the congruence 1.2.7.8. This proves the result 1.2.7.4.

By assumption $f(\alpha(X)) - f(\beta(X)) \in I^r R[\![X]\!]$, we have that

$$\delta(X) := f^{-1}(f(\beta(X)) - f(\alpha(X))) \in I^r R[\![X]\!].$$

Then

$$\begin{aligned} f(\beta(X)) &= f(\delta(X)) + f(\alpha(X)) \\ \beta(X) &= f^{-1}(f(\delta(X)) + f(\alpha(X))) \end{aligned}$$

It is straightforward to prove that $f^{-1}(f(X) + f(Y))$ is a formal group law, then

$$\beta(X) = f^{-1}(f(\delta(X)) + f(\alpha(X))) = \alpha(X) + \delta(X) G(\delta(X), \alpha(X)). \tag{1.2.7.9}$$

for some $G(X, Y) \in R[\![X, Y]\!]$. Since $\delta(X) \in I^r R[\![X]\!]$, then $\beta(X) \equiv \alpha(X) (\operatorname{mod} I^r R[\![X]\!])$.

□

1.3 Universal formal group laws

In this section, we discuss the construction of universal formal group laws and their applications.

Definition 1.3.1 ([6, Chap. 1, Definition 1.5.2])**.** A one-dimensional formal group law $F(X, Y)$ over L is called a universal formal group law if and only if, for any formal group law $G(X, Y)$ over R, there is a unique ring homomorphism $\phi : L \to R$ such that $\phi_* F(X, Y) = G(X, Y)$.

Remark 1.3.2. The ring L is uniquely determined up to isomorphism. Indeed, let $F(X,Y)$ and $G(X,Y)$ be universal formal group laws over rings L and L', respectively. Then there are ring homomorphisms $\phi : L \to L'$ and $\psi : L' \to L$ such that $\phi_* F(X,Y) = G(X,Y)$ and $\psi_* G(X,Y) = F(X,Y)$. By uniqueness, we can deduce that $\phi\psi = \mathrm{id}_{L'}$ and $\psi\phi = \mathrm{id}_L$.

We can show that there exists a universal formal group law over some ring L. Let $\tilde{L} = \mathbb{Z}[C_{i,j}]$, where $C_{i,j}$ are indeterminates and $i,j = 1,2,\ldots$. Consider the power series $F_C(X,Y) = X + Y + \sum_{i,j} C_{i,j} X^i Y^j$. We write

$$F_C(F_C(X,Y),Z) - F_C(X,F_C(Y,Z)) = \sum_{i,j,k} P_{i,j,k}(C) X^i Y^j Z^k$$

where $P_{i,j,k}(C)$ is a polynomial in the variables $C_{i,j}$. Let I be the ideal of $\tilde{L}$ generated by $C_{i,j} - C_{j,i}$ and $P_{i,j,k}(C)$. Let $L = \tilde{L}/I$ and $F(X,Y) = p_* F_C(X,Y)$, where $p : \tilde{L} \to L$ is the natural quotient map. Therefore, given a formal group law $F(X,Y)$ over a ring R, if we denote $F(X,Y) = a_{ij} X^i Y^j$, then there is a unique ring homomorphism $\phi : L \to R$ such that $\phi(C_{ij}) = a_{ij}$ and $\phi_* F_C(X,Y) = F(X,Y)$. The uniqueness follows from the equation $\phi_* F_C(X,Y) = F(X,Y)$.

However, we know nothing about the structure of the ring L and $F(X,Y)$ above. In fact, we can prove that L is isomorphic to $\mathbb{Z}[U] := \mathbb{Z}[U_2, U_3, \ldots]$, where U_i are indeterminates, and $F(X,Y) \in \mathbb{Z}[U][\![X]\!]$ is of the form $f^{-1}(f(X)+f(Y))$ for some $f(X) \in \mathbb{Q}[U][\![X]\!]$. The integrality of the coefficients of $F(X,Y)$ then follows from the functional equation lemma.

Remark 1.3.3. There is a functor from the category of commutative rings to **Type**, which maps a commutative ring R to formal group laws over R, and maps a ring homomorphism $\phi : R \to S$ to the pushforward map from formal group laws over R to formal group laws over S, as defined in Definition 1.1.11. This is a covariant functor and is corepresented by the ring L and $F_C(X,Y)$. To be specific, there is a bijection between $\mathcal{H}om_{Ring}(L,R)$ and formal group laws over R.

1.4 Commutativity theorem

In this section, we prove that 'almost all' one-dimensional formal groups are commutative. More precisely,

Theorem 1.4.1 ([6, Chap. 1 Theorem 6.1])**.** *Let R be a ring. Then every one-dimensional formal group law over R is commutative if and only if R has no nonzero element a that is both nilpotent and $\mathbb{Z}$-torsion.*

Before proving the theorem, we first discuss some lemmas.

Lemma 1.4.2 ([6, Chap. 1 Lemma 6.2.1])**.** *Let $F(X,Y)$ be a non-commutative one-dimensional formal group law over a ring R. Then there is a nonzero homomorphism from $F(X,Y)$ into $\mathbf{G}_a(X,Y)$ or into $\mathbf{G}_m(X,Y)$.*

Proof. For simplicity, we denote $F(X,Y)$ by $X +_F Y$ and the inverse $i(X)$ by $-_F X$. Let $H(X,Y) = X +_F Y + (-_F X)$. We have $H(0,Y) = Y$. Since $F(X,Y)$ is non-commutative, we have

$$H(X,Y) = Y + \sum_{i=1}^{\infty} r_n(X) Y^n \tag{1.4.2.1}$$

with $r_n(X) \neq 0$ for some n. Otherwise, $H(X,Y) = Y$, which means that $F(X,Y)$ is commutative. Since $H(0,Y) = Y$, we have $r_n(0) = 0$ for all n. By associativity of $F(X,Y)$, we have

$$H(X_1, H(X_2, Y)) = H(X_1 +_F X_2, Y) \tag{1.4.2.2}$$

Let i be the least integer such that $r_i(X) \neq 0$.

For the case $i = 1$, we have

$$H(X,Y) \equiv (1 + r_1(X))Y \pmod{Y^2}$$

Combining this with equation 1.4.2.2, we have

$$H(X_1, H(X_2, Y)) \equiv (1 + r_1(X_1))H(X_2, Y) \equiv (1 + r_1(X_1))(1 + r_1(X_2))Y \pmod{Y^2} \tag{1.4.2.3}$$

$$H(X_1 +_F X_2, Y) \equiv (1 + r_1(X_1 +_F X_2))Y \pmod{Y^2} \tag{1.4.2.4}$$

Therefore, $(1 + r_1(X_1))(1 + r_1(X_2)) = 1 + r_1(X_1 +_F X_2)$, i.e.

$$r_1(X_1 +_F X_2) = r_1(X_1) + r_1(X_2) + r_1(X_1)r_1(X_2). \tag{1.4.2.5}$$

This means $r_1(X)$ is a formal group homomorphism from F to $\mathbf{G}_m$.

For the case $i > 1$, we have

$$H(X,Y) \equiv Y + r_i(X)Y^i \pmod{Y^{i+1}}$$

According to equation 1.4.2.2, we have

$$H(X_1, H(X_2, Y)) \equiv H(X_2, Y) + r_i(X_1)H(X_2, Y)^i \equiv Y + r_i(X_2)Y^i + r_i(X_1)Y^i \pmod{Y^i}. \tag{1.4.2.6}$$

$$H(X_1 +_F X_2, Y) \equiv Y + r_i(X_1 +_F X_2)Y^i \pmod{Y^{i+1}}. \tag{1.4.2.7}$$

Therefore, $r_i(X_1 +_F X_2) = r_i(X_1) + r_i(X_2)$, which means $r_i(X)$ is a formal group homomorphism from F to $\mathbf{G}_a$. □

Lemma 1.4.3 ([6, Chap. 1 Lemma 6.2.11])**.** *Let R be an integral domain and let F, F' be one-dimensional formal group laws over R. Suppose there exists a homomorphism $\alpha(X) : F(X,Y) \to F'(X,Y)$ and $F'(X,Y)$ is commutative. Then $F(X,Y)$ is commutative.*

Proof. Let r be the order of $\alpha(X) = \sum_{n=1}^{\infty} a_n X^n$, i.e. the least n such that a_n does not equal zero. Let $C(X,Y) = X +_F Y +_F (-_F X) +_F (-_F Y)$ be the commutator of X and Y with respect to F.

We can prove that a formal group law F is commutative if and only if the commutator $C(X,Y)$ with respect to F is zero.

Since F' is commutative, we have

$$\alpha(C(X,Y)) = 0.$$

Suppose F is non-commutative, i.e. $C(X,Y) \neq 0$. Let $C_n(X,Y)$ be the homogeneous component in $C(X,Y)$ of degree n, and let m be the least integer such that $C_m(X,Y)$ is non-zero. We have

$$\alpha(C(X,Y)) \equiv a_r C_m(X,Y)^r \pmod{\deg(mr+1)}$$

Since $a_r \in R$ and R is an integral domain, this term is non-zero, which contradicts the assumption that $\alpha(C(X,Y)) = 0$. □

Proof of Commutativity theorem 1.4.1. First, we prove the necessity of the condition on R.

Let $0 \neq a \in R$ be an element of R that is nilpotent and $\mathbb{Z}$-torsion. Then there is a formal group law $F(X,Y)$ over R such that $F(X,Y)$ is non-commutative.

Let n be the least integer such that $na = 0$. Then $n > 1$ and there exists a prime number p such that $p \mid n$. Let $b = \frac{n}{p}a$. Then $b \neq 0$ and $pb = 0$, since n is minimal. Let m be the least integer such that $b^m = 0$ and $c = b^{m-1}$. Therefore, this c satisfies $c \neq 0, c^2 = 0, pc = 0$. Denote $F(X,Y) = X + Y + cXY^p$. Then

$$F(F(X,Y),Z) = X + Y + Z + c(XY^p + XZ^p + YZ^p) = F(X, F(Y,Z))$$

Therefore, $F(X,Y)$ is a non-commutative formal group law over R.

Secondly, we prove sufficiency for the case where R is an integral domain. Let $F(X,Y)$ be a formal group law over R, where R is an integral domain. Suppose that $F(X,Y)$ is non-commutative. Then, by Lemma 1.4.2, there is a nonzero formal group homomorphism from $F(X,Y)$ to $\mathbf{G}_a(X,Y)$ or $\mathbf{G}_m(X,Y)$. Since $\mathbf{G}_a(X,Y)$ and $\mathbf{G}_m(X,Y)$ are commutative formal group laws, Lemma 1.4.3 implies that $F(X,Y)$ is commutative. This leads to a contradiction.

Thirdly, the proof of sufficiency for the case where R is $\mathbb{Z}$-torsion free follows directly from computation on the associativity condition of a formal group law. There is a proof later in Corollary 1.6.4. It can also be found in [6, Chap. 1 Lemma 6.1.3].

Now we prove the sufficiency of the condition on R for theorem 1.4.1.

Given a prime ideal $\mathfrak{p}$ in R, consider the ring homomorphism $\phi : R \to R/\mathfrak{p}$. Since $R/\mathfrak{p}$ is an integral domain, then $\phi_* F(X,Y)$ is commutative. If we denote $F(X,Y) = \sum_{i,j} a_{ij} X^i Y^j$, then it follows that $\phi_*(a_{ij} - a_{ji}) = 0 \in R/\mathfrak{p}$, which is equivalent to $a_{ij} - a_{ji} \in \mathfrak{p}$. Since this holds for all prime ideals $\mathfrak{p}$, we have

$$a_{ij} - a_{ji} \in \bigcap_{\mathfrak{p}\ prime} \mathfrak{p}$$

Therefore, for any i, j, $a_{ij} - a_{ji}$ is nilpotent in R.

Consider the ring homomorphism $\phi_\infty : R \to R \otimes_{\mathbb{Z}} \mathbb{Q}$. Since $R \otimes_{\mathbb{Z}} \mathbb{Q}$ is $\mathbb{Z}$-torsion free, then $\phi_{\infty *} F(X,Y)$ is commutative. Therefore, $\phi_\infty(a_{ij} - a_{ji}) = 0 \in R \otimes_{\mathbb{Z}} \mathbb{Q}$, which means $a_{ij} - a_{ji}$ is $\mathbb{Z}$-torsion. Since there is no nonzero element in R that is nilpotent and $\mathbb{Z}$-torsion at the same time, it follows that for any i, j, $a_{ij} - a_{ji} = 0$, which implies $F(X,Y)$ is commutative.

□

1.5 Invariant differential

Let F be a formal group law over R. In this setting, a differential form is of the form $P(X)dX$ for some $P(X) \in R[\![X]\!]$. In this section, we discuss those differential forms that respect the group structure of F.

Definition 1.5.1 ([11, Chap. 4 Definition 4.1])**.** An invariant differential on a formal group law F over R is a differential form $\omega(X) = P(X)dX$ with $P(X) \in R[\![X]\!]$ satisfying

$$\omega(X) \circ F(X,Y) = \omega(X).$$

To be precise, $P(F(X,Y))F_X(X,Y) = P(X)$, where $F_X(X,Y)$ is the partial derivative of F with respect to its first variable. An invariant differential is called normalized if $P(0) = 1$.

Recall that a power series f over R is invertible in the ring of power series if and only if its constant coefficient is invertible in R.

Proposition 1.5.2 ([11, Chap. 4 Proposition 4.2])**.** *Let F be a formal group law over R. Then there exists a unique normalized invariant differential on F. It is given by the formula*

$$\omega(X) = F_X(0, X)^{-1} dX.$$

Moreover, every invariant differential on F can be written as $a\omega$ for some $a \in R$.

Proof. Write $\omega(X) = P(X)dX$. Since ω is invariant, we must have

$$P(F(X, Y))F_X(X, Y) = P(X).$$

Let $X = 0$ in the above equation and use the fact that $F(0, Y) = Y$. Then we have

$$P(Y)F_X(0, Y) = P(0).$$

Since $F_X(0, Y)$ is of the form $(1 + \cdots)$, $F_X(0, Y)$ is invertible in $R[\![Y]\!]$. It follows that $P(X)$ is completely determined by the value $P(0)$. Therefore, it is enough to verify that

$$\omega = F_X(0, X)^{-1} dX$$

is a normalized invariant differential.

It is equivalent to show

$$F_X(0, F(X, Y))^{-1} F_X(X, Y) = F_X(0, X)^{-1}.$$

Consider the associativity condition $F(F(X, Y), Z) = F(X, F(Y, Z))$ and differentiate it with respect to X. Then we have

$$F_X(X, F(Y, Z)) = F_X(F(X, Y), Z)F_X(X, Y).$$

Taking $X = 0$ above, we have

$$F_X(0, F(Y, Z)) = F_X(Y, Z)F_X(0, Y).$$

This is exactly what we want. It is normalized since $F_X(0, X)$ is of the form $(1 + \cdots)$. □

Example 1.5.3. The normalized invariant differential on $\mathbf{G}_a$ is dX, and the normalized invariant differential on $\mathbf{G}_m$ is $(1 + X)^{-1}dX = \sum_{n=0}^{\infty}(-1)^n X^n dX$.

Corollary 1.5.4 ([11, Chap. 4 Corollary 4.3])**.** *Let F, G be formal group laws over R with normalized invariant differentials ω_F and ω_G, respectively. Let f be a homomorphism from F to G. Then*

$$\omega_G \circ f = f'(0)\omega_F.$$

where $f'(X)$ is the formal derivative of $f(X)$.

Proof. According to the definition, we have

$$\omega_G \circ f(F(X, Y)) = \omega_G \circ G(f(X), f(Y)) = \omega_G \circ f(X).$$

Then $\omega_G \circ f$ is an invariant differential on F. It follows that $\omega_G \circ f$ can be expressed as $a\omega_F$ for some $a \in R$. By checking the coefficient of X, we can deduce that $a = f'(0)$. □

Corollary 1.5.5 ([11, Chap. 4 Corollary 4.4])**.** *Let F be a formal group law over R and let p be a prime number. Then there are power series $f(X), g(X)$ over R with zero constant coefficient such that*

$$[p]_F(X) = pf(X) + g(X^p).$$

Proof. Let ω be the normalized invariant differential on F. We have discussed before that $[p](X) = pX + \cdots$. Then by Corollary 1.5.4, we have

$$\omega \circ [p] = p\omega.$$

If we denote $\omega \circ [p](X) = P(X)[p]'(X)dX$, then $P(X)$ is of the form $(1 + \cdots)$ since ω is a normalized invariant differential on F. Since $P(X)$ is invertible, we have

$$[p]'(X) \in pR[\![X]\!].$$

Then each coefficient a_n of $[p](X)$ satisfies either $p \mid n$ or $a_n \in pR$. □

Remark 1.5.6. Assume that R is a ring of characteristic p. Then the p-series $[p]$ has the form $[p](X) = g(X^p)$ for some $g(X) \in R[\![X]\!]$.

1.6 Logarithm

In this section, we discuss the formal logarithm of a formal group law. By integrating an invariant differential form, we expect to obtain a homomorphism to the additive group. However, integration introduces denominators. In the case of characteristic 0, or over torsion-free rings, things work as expected.

From now on, we fix a torsion-free ring R and let $K = R \otimes \mathbb{Q}$.

Definition 1.6.1 ([11, Chap. 4 Definition 5.1])**.** Let F be a formal group law over R, and let $\omega(X)$ be its normalized invariant differential. Write $\omega(X) = (1 + a_1X + a_2X^2 + \cdots)dX$. Then the formal logarithm of F is defined by

$$\log_F(X) := \int \omega = X + \frac{a_1}{2}X^2 + \frac{a_2}{3}X^3 + \cdots \in K[\![X]\!]$$

Since the coefficient of X in $\log_F(X)$ is 1, which is invertible in K, $\exp_F(X)$ is defined to be the unique power series over K satisfying

$$\exp_F(\log_F(X)) = X.$$

Example 1.6.2. The normalized invariant differential on $\mathbf{G}_a$ is dX, so we have

$$\log_{\mathbf{G}_a}(X) = X, \ \exp_{\mathbf{G}_a}(X) = X.$$

The normalized invariant differential on $\mathbf{G}_m$ is $(1 + X)^{-1}dX$, so we have

$$\log_{\mathbf{G}_m}(X) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}X^n}{n}$$

$$\exp_{\mathbf{G}_m}(X) = \sum_{n=1}^{\infty} \frac{X^n}{n!}$$

As expected, the formal logarithm of F is a homomorphism from F to $\mathbf{G}_a$. However, since $\log_F$ introduces denominators, this is no longer a homomorphism over R; it is a homomorphism over K. Moreover, it is an isomorphism over K.

Proposition 1.6.3 ([11, Chap. 4 Proposition 5.2])**.** *Let F be a formal group law over R. Then $\log_F(X)$ is a formal group isomorphism from F to $\mathbf{G}_a$ over K.*

Proof. Let $\omega(X)$ be the normalized invariant differential for F. Then we have

$$\omega \circ F = \omega.$$

Integrating this, we have

$$\log_F(F(X,Y)) = \log_F(X) + C(Y).$$

where $C(Y) \in K[\![Y]\!]$. Let $X = 0$ in the above equation. Then we have

$$C(Y) = \log_F(0) + C(Y) = \log_F(F(0,Y)) = \log_F(Y).$$

Hence,

$$\log_F(F(X,Y)) = \log_F(X) + \log_F(Y).$$

This proves that $\log_F(X)$ is a homomorphism from F to $\mathbf{G}_a$. Furthermore, since $\exp_F$ is its inverse, $\exp_F$ is a homomorphism from $\mathbf{G}_a$ to F. Therefore, $\log_F$ is an isomorphism. □

Corollary 1.6.4. *Let R be a torsion-free ring. Then every formal group law over R is commutative.*

Proof. The assumption that R is torsion free guarantees the existence of $\log_F$ and $\exp_F$. Then

$$F(X,Y) = \exp_F(\log_F(X) + \log_F(Y)).$$

Therefore, we have $F(X,Y) = F(Y,X)$. □

Further discussion of the coefficients of $\log_F$ and $\exp_F$ can be found in [11].

1.7 Height

In this section, we discuss the definition and properties of the height of a formal group law. Let $F(X,Y)$ be a commutative formal group law over a ring R, fixed throughout this section.

From now on, we fix a prime number p and assume that R has characteristic p.

Definition 1.7.1 (Height of homomorphism [11, Chap. 4 Definition 7.1])**.** Let F, G be formal group laws over R, and let $f \in R[\![X]\!]$ be a homomorphism from $F(X,Y)$ to $G(X,Y)$. Then the height of f, denoted by $\mathrm{ht}(f)$, is the largest number h such that

$$f(X) = g(X^{p^h})$$

for some power series $g(X)$ over R. In particular, if $f = 0$, then we say $\mathrm{ht}(f) = \infty$.

Remark 1.7.2. Let m be a natural number prime to p, then $\mathrm{ht}([m]) = 0$, since

$$[m](X) = mX + \cdots$$

Definition 1.7.3 (Height of formal group)**.** The height of a formal group law F is defined to be the height of the p-series $[p]_F(X)$, denoted by $\mathrm{ht}(F)$. More precisely, let v_n be the p^n-th coefficient of the p-series $[p](X)$. We say $F(X,Y)$ has height n if $v_i = 0$ for all $i < n$ and $v_n \neq 0$ in R. In particular, we say $F(X,Y)$ has infinite height if $[p](X) = 0$.

Remark 1.7.4. From the remark above, we have $v_0 = p$. Then $F(X,Y)$ has height ≥ 1 if and only if $p = 0$ in R, which is equivalent to saying either R has characteristic p or R is trivial. Also, $F(X,Y)$ has height exactly 0 if and only if p is invertible in R.

Example 1.7.5. The p-series for the multiplicative formal group $\mathbf{G}_m(X,Y) = (1+X)(1+Y)-1$ is $[p](X) = (1+X)^p - 1 = X^p$, so $\mathbf{G}_m(X,Y)$ has height exactly 1. For the additive formal group law $\mathbf{G}_a(X,Y)$, the p-series is $[p](X) = pX = 0$, so $\mathbf{G}_a(X,Y)$ has infinite height.

Proposition 1.7.6. *Let $F(X,Y)$ and $G(X,Y)$ be formal group laws over a ring R. Then $F(X,Y)$ and $G(X,Y)$ have the same height.*

Proof. Let $g(X,Y)$ be a formal group isomorphism from $F(X,Y)$ to $G(X,Y)$, and let $[p]_F(X)$ and $[p]_G(X)$ be the p-series for $F(X,Y)$ and $G(X,Y)$, respectively. Note that

$$[p]_G(X) = (g \circ [p]_F \circ g^{-1})(X)$$

It follows that the heights of $F(X,Y)$ and $G(X,Y)$ are the same. □

Remark 1.7.7. The additive and multiplicative formal groups are isomorphic over a ring R of characteristic p.

Proposition 1.7.8 ([11, Chap. 4 Proposition 7.3])**.** *Let F, G, H be formal group laws over R, and let*

$$F \xrightarrow{f} G \xrightarrow{g} H$$

be a chain of homomorphisms defined over R. Then

$$\mathrm{ht}(g \circ f) = \mathrm{ht}(f) + \mathrm{ht}(g)$$

Proof. Denote $f(X) = f_1(X^{p^{\mathrm{ht}(f)}})$ and $g(X) = g_1(X^{p^{\mathrm{ht}(g)}})$. Then

$$(g \circ f)(X) = g_1(f_1(X^{p^{\mathrm{ht}(f)}})^{\mathrm{ht}(g)}) = g_1(\tilde{f}_1(X^{\mathrm{ht}(f)+\mathrm{ht}(g)}))$$

where $\tilde{f}_1(X) = \mathrm{Frob}_*^{\mathrm{ht}(g)} f_1(X)$, namely raising each coefficient of f_1 to the $p^{\mathrm{ht}(g)}$-th power. □

Proposition 1.7.9 ([11, Chap. 4 Proposition 7.2])**.** *Let F, G be formal group laws over R, and let f be a homomorphism from F to G defined over R. Then*

1. *If $f'(0) = 0$, then $f(X) = f_1(X^p)$ for some $f_1(X) \in R[\![X]\!]$. That is, $\mathrm{ht}(f) \geq 1$.*
2. *Write $f(X) = g(X^{p^{\mathrm{ht}(f)}})$. Then $g'(0) \neq 0$.*

Proof. The proof of this proposition needs results discussed in the section on invariant differentials.

For the first part of this proposition, let ω_F, ω_G be the normalized invariant differentials of F and G, respectively. According to Corollary 1.5.4, we have

$$\begin{aligned} 0 &= f'(0)\omega_F \\ &= \omega_G \circ f \end{aligned}$$

If we write $\omega_G(X) = P_G(X)dX$, then we have

$$\omega_G \circ f(X) = \omega_G(f(X)) = P_G(f(X))f'(X)dX.$$

Then $f'(X) = 0$, since $P_G(X)$ is of the form $(1 + \cdots)$.

For the second part of this proposition, we first denote $q = p^{\mathrm{ht}(f)}$ and $F_q = \phi_* F$, where ϕ is a ring endomorphism, mapping $a \mapsto a^q$. Then F_q is also a formal group law. Now we claim that $g(X)$ is a homomorphism from F_q to G. We prove this claim by the following calculation:

$$\begin{aligned} g(F_q(X,Y)) &= g(F(X,Y)^q) \\ &= g(X^q) \circ F \\ &= f(X) \circ F \\ &= g \circ F \end{aligned}$$

The first equation is guaranteed by the characteristic of R. If $g'(0) \neq 0$, then by the first assertion of the proposition, there is a $g_1(X)$ such that $g(X) = g'(X^p)$. It follows that $f(X) = g_1(X^{p^{\mathrm{ht}(f)+1}})$, which contradicts the definition of the height of f. $\square$

Theorem 1.7.10 (Lazard [8, Theorem IV])**.** *Let k be an algebraically closed field of characteristic p. Then two formal group laws $F(X,Y), G(X,Y)$ over k are isomorphic if and only if they have the same height.*

2 Lubin-Tate Theory

2.1 Definition and Properties

Let K be a field complete with respect to a discrete valuation, with finite residue field. For instance, K is a nonarchimedean local field. Let $\mathcal{O}_K$ be the ring of integers of K, let q be the cardinality of the residue field, and let π be a uniformizer of $\mathcal{O}_K$.

Definition 2.1.1 ([2, Chap. 6 Definition 3.3])**.** Let $\mathcal{F}_\pi$ denote the set of power series $f(X) \in \mathcal{O}_K[\![X]\!]$ which satisfy the two conditions:

$$f(X) \equiv \pi X \pmod{\deg 2} \tag{2.1.1.1}$$

$$f(X) \equiv X^q \pmod{\pi} \tag{2.1.1.2}$$

Lemma 2.1.2 ([2, Chap. 6 Proposition 5])**.** *Let $f(X)$ and $g(X)$ be elements of $\mathcal{F}_\pi$ and let $L(X_1, \ldots, X_n) = \sum_{i=1}^n a_i X_i$ be a linear form with coefficients in $\mathcal{O}_K$.*

Then there exists $F(X_1, \ldots, X_n)$ with coefficients in $\mathcal{O}_K$ such that

$$F(X_1, \ldots, X_n) \equiv L(X_1, \ldots, X_n) (\mathrm{mod}\ \deg 2) \tag{2.1.2.1}$$

$$f(F(X_1, \ldots, X_n)) = F(g(X_1), \ldots, g(X_n)) \tag{2.1.2.2}$$

Proof. We use induction to prove that for all $k \in \mathbb{N}$, there exists a unique multivariate polynomial $F_k(X_1, \ldots, X_n)$ of degree k and

$$F_k(X_1, \ldots, X_n) \equiv L(X_1, \ldots, X_n) (\mathrm{mod} \deg 2)$$
$$f(F_k(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) (\mathrm{mod} \deg(k+1))$$

The existence and uniqueness of $F(X_1, \ldots, X_n)$ then follow from this.

For $k = 1$, we take $F_1(X_1, \ldots, X_n) = L(X_1, \ldots, X_n)$.

Now assume that the result holds for k. Write $f(F_k(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + E_{k+1} (\mathrm{mod} \deg(k+2))$. According to the induction hypothesis, E_{k+1} is homogeneous of degree $k+1$.

On the other hand, since the uniqueness of $F_k(X_1, \ldots, X_n)$, then $F_{k+1}(X_1, \ldots, X_n)$ should be of the form $F_{k+1}(X_1, \ldots, X_n) = F_k(X_1, \ldots, X_n) + \phi_{k+1}(X_1, \ldots, X_n)$, where $\phi_{k+1}(X_1, \ldots, X_n)$ is homogeneous of degree $k+1$, due to the definition of $\mathcal{F}_\pi$. Then we have

$$f(F_{k+1}(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + \pi \phi_{k+1}(X_1, \ldots, X_n) (\mathrm{mod} \deg(k+2))$$
$$F_{k+1}(g(X_1), \ldots, g(X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + \pi^{k+1} \phi_{k+1}(X_1, \ldots, X_n) (\mathrm{mod} \deg(k+2))$$

The last congruence follows from the fact that ϕ_{k+1} is homogeneous of degree $k+1$, i.e. $\phi_{k+1}(g(X_1), \ldots, g(X_n)) = \pi^{k+1} \phi_{k+1}(X_1, \ldots, X_n)$. Therefore, we conclude that

$$E_{k+1} + (\pi - \pi^{k+1}) \phi_{k+1}(X_1, \ldots, X_n) = 0 \tag{2.1.2.3}$$

Then $\phi_{k+1}(X_1, \ldots, X_n) = -E_{k+1}/(\pi - \pi^{k+1})$. □

Definition 2.1.3. For any $f \in \mathcal{F}_\pi$, we let $F_f(X,Y)$ be the unique solution of

$$F_f(X,Y) \equiv X + Y (\text{mod } \deg 2) \tag{2.1.3.1}$$
$$f(F_f(X,Y)) = F_f(f(X), f(Y)) \tag{2.1.3.2}$$

Theorem 2.1.4 (Lubin-Tate Formal Group Laws [2, Chap. 6 Proposition 1])**.** *For power series $f \in \mathcal{F}_\pi$, we have*

$$F_f(X,Y) = F_f(Y,X) \tag{2.1.4.1}$$
$$F_f(F_f(X,Y),Z) = F_f(X, F_f(Y,Z)) \tag{2.1.4.2}$$

As a result, $F_f(X,Y)$ is a commutative formal group law over $\mathcal{O}_K$.

Proof. For equation 2.1.4.1, let $G(X,Y) = F(Y,X)$, then $G(X,Y) \equiv X + Y (\text{mod} \deg 2)$ and

$$f(G(X,Y)) = f(F(Y,X)) = F(f(Y), f(X)) = G(f(X), f(Y))$$

Therefore, by the uniqueness in Lemma 2.1.2, $F(X,Y) = G(X,Y) = F(Y,X)$.

For equation 2.1.4.2, let $G_1(X,Y,Z) = F_f(F_f(X,Y),Z)$ and $G_2(X,Y,Z) = F_f(X, F_f(Y,Z))$, and we can check that $G_i(X,Y,Z) \equiv X + Y + Z (\text{mod} \deg 2)$ and

$$f(G_i(X,Y,Z)) = G_i(f(X), f(Y), f(Z))$$

Then by the uniqueness in Lemma 2.1.2, we conclude the desired equation. □

Definition 2.1.5. For each $a \in \mathcal{O}_K$ and $f, g \in \mathcal{F}_\pi$, let $[a]_{f,g}(X)$ be the unique solution of

$$[a]_{f,g}(X) \equiv aX (\text{mod} \deg 2) \tag{2.1.5.1}$$
$$f([a]_{f,g}(X)) = [a]_{f,g}(g(X)) \tag{2.1.5.2}$$

In particular, denote $[a]_f$ instead of $[a]_{f,f}$ to simplify the notation.

Theorem 2.1.6 ([2, Chap. 6 Proposition 2])**.** *For each $a \in \mathcal{O}_K$ and $f, g \in \mathcal{F}_\pi$, the power series $[a]_{f,g}(X)$ is a formal group homomorphism from $F_g(X,Y)$ to $F_f(X,Y)$, i.e. $[a]_{f,g}F_g(X,Y) = F_f([a]_{f,g}(X), [a]_{f,g}(Y))$.*

Proof. For simplicity, we denote $[a]_{f,g}$ as h. We have $h(F_g(X,Y)) \equiv aX + aY (\text{mod} \deg 2)$, $F_f(h(X), h(Y)) \equiv aX + aY (\text{mod} \deg 2)$ and

$$h \circ F_g(g(X), g(Y)) = h \circ g(F_g(X,Y)) = f \circ h(F_f(X,Y))$$
$$F_f(h(g(X), h(g(Y)))) = F_f(f(h(X)), f(h(Y))) = f \circ F_f(h(X), h(Y))$$

Then the result follows from uniqueness in Lemma 2.1.2. □

Theorem 2.1.7 ([9, Theorem 1])**.** *Let $a, b \in \mathcal{O}_K$, $f, g, h \in \mathcal{F}_\pi$, then $[a]_{f,g}([b]_{g,h}(X)) = [ab]_{f,h}(X)$.*

Proof. We have $[a]_{f,g}([b]_{g,h}(X)) \equiv abX (\text{mod} \deg 2)$ and

$$f \circ ([a]_{f,g} \circ [b]_{g,h}) = [a]_{f,g} \circ (g \circ [b]_{g,h}) = [a]_{f,g} \circ [b]_{g,h} \circ h$$

Then we have $[ab]_{f,h}$ and $[a]_{f,g} \circ [b]_{g,h}$ are solutions satisfying these conditions. By the uniqueness of Lemma 2.1.2, they coincide. □

Theorem 2.1.8 ([9, Theorem 1]). *Let $a, b \in \mathcal{O}_K$, $f, g \in \mathcal{F}_\pi$, then $F_f([a]_{f,g}(X), [b]_{f,g}(X)) = [a+b]_{f,g}(X)$.*

Proof. Notice that $F_f([a]_{f,g}(X), [b]_{f,g}(X)) \equiv (a+b)X (\operatorname{mod} \deg 2)$ and

$$f(F_f([a]_{f,g}(X), [b]_{f,g}(X))) = F_f(f \circ [a]_{f,g}(X), f \circ [b]_{f,g}(X)) = F_f([a]_{f,g} \circ f(X), [b]_{f,g} \circ f(X))$$

Therefore, by the uniqueness in Lemma 2.1.2, we have $F_f([a]_{f,g}(X), [b]_{f,g}(X)) = [a+b]_{f,g}(X)$. □

Theorem 2.1.9 ([2, Chap. 6 Proposition 4]). *Let $f, g \in \mathcal{F}_\pi$, then F_f is isomorphic to F_g, and there is a unique strong isomorphism from F_f to F_g. This means that the Lubin-Tate formal group F_f is independent of $f \in F_\pi$, and depends only on π.*

Proof. Take a unit $a \in \mathcal{O}_K$, then $[a]_{f,g} \circ [a^{-1}]_{g,f} = [1]_f$ and $[a^{-1}]_{g,f} \circ [a]_{f,g} = [1]_g$. And the unique strong isomorphism is $[1]_{f,g}$. □

Remark 2.1.10. Let L be an algebraic extension of K, let $\mathfrak{m}_L$ be the maximal ideal in the ring of integers of L. Given elements $x_1, \ldots, x_n \in \mathfrak{m}_L$ and a formal series $G(X_1, \ldots, X_n) \in \mathcal{O}_K[\![X_1, \ldots, X_n]\!]$, the series $G(x_1, \ldots, x_n)$ converges to an element in $\mathfrak{m}_L$ if the constant term of $G(X_1, \ldots, X_n)$ is zero. By above theorem, for each power series $f \in \mathcal{F}_\pi$ this determines an $\mathcal{O}_K$-module structure on the set $\mathfrak{m}_L$. To be precisely, addition and scalar multiplication are defined by

$$x + y = F_f(x, y) \tag{2.1.10.1}$$

$$ax = [a]_f(x) \tag{2.1.10.2}$$

2.2 Applications

Fix an algebraic closure $\overline{K}$ of K, and let K^{sep} be a separable closure. Given a $f \in \mathcal{F}_\pi$, for each integer $n \geq 1$, let $f^n(X)$ be n-th iterate of $f(X)$. Let $\Lambda_{f,n}$ be the set of all roots of $f^n(X)$ in $\overline{K}$ and let $K(\Lambda_{f,n})$ be the field generated over K by these elements. Moreover, we have the fact that for any $f, g \in \mathcal{F}_\pi$, $\lambda \in \Lambda_{f,m}$ if and only if $[1]_{g,f}(\lambda) \in \Lambda_{g,m}$. This means that the field $K(\Lambda_{f,n})$ is independent of f and depends only on uniformizer π. Therefore, we denote $K(\Lambda_{f,n})$ as $L_{\pi,n}$. The same argument holds for $K(\Lambda_f)$. Additionally, let $\Lambda_f = \bigcup_{n=1}^{\infty} \Lambda_{f,n}$, $L_\pi = K(\Lambda_f)$, $G_{\pi,n} = \mathrm{Gal}(L_{\pi,n}/K)$, $G_\pi = \varprojlim G_{f,n}$.

According to the remark 2.1.10, F_f defines a $\mathcal{O}_K$ structure on $\mathfrak{m}_{K^{sep}}$ and we denote this $\mathcal{O}_K$-module as $M_f(K^{sep})$. Then $\Lambda_{f,n}$ is those elements in $\lambda \in \mathfrak{m}_{K^{sep}}$ such that $\pi^n \lambda = 0$.

Theorem 2.2.1 ([2, Chap. 6 Theorem 3]). *The $\mathcal{O}_K$-module $\Lambda_{f,n}$ is isomorphic to $\mathcal{O}_K/\pi^n\mathcal{O}_K$ and $\mathcal{O}_K$-module Λ_f is isomorphic to $K/\mathcal{O}_K$.*

In order to prove this theorem, we need a result about divisible modules over a PID.

Theorem 2.2.2 (Structure theorem of divisible module over PID [10]). *Let R be a principal ideal domain and K be its field of fractions. Then every divisible R-module is isomorphic to a direct sum of K and copies of the injective hulls of the simple module $R/(p)$, varying over prime elements $p \in R$.*

Let M be a divisible R-module, then M can be written as follows:

$$M \cong (\bigoplus_{i \in I} K) \oplus (\bigoplus_{p \text{ prime}} \bigoplus_{j \in J_p} E(R/(p)))$$

- *I is an index set measuring how many copies of torison free part in M are needed.*
- *p varies over all prime elements in R.*
- *J_p is an index set measuring how many copies of p-primary torison part $E(R/(p))$ are needed for each p.*
- *$E(R/(p))$ is the injective hull of $R/(p)$. Over PID, this is isomorphic to $K/R_{(p)}$.*

Proof. The proof of the structure theorem for divisible modules can be found in Chapter 3 of [7] or Theorem 2.5 and Proposition 3.1 in [10]. □

Proof of theorem 2.2.1. The first claim follows by the fact that $\Lambda_{f,n}$ is the π^n-torsion of $\mathcal{O}_K$-module $M_f(K^{sep})$.

According to theorem 2.1.9, we are free to choose $f \in \mathcal{F}_\pi$ as we please. We take $f(X) = \pi X + X^q$. Since $f'(X) = qX^{q-1} + \pi \neq 0$, then $\Lambda_{f,1}$ consists of q distinct roots of the $X^q + \pi X = 0$. For any given $\alpha \in \mathfrak{m}_{K^{sep}}$, since $\pi X + X^q = \alpha$ is separable, then $\Lambda_{f,1} \subset \mathfrak{m}_{K^{sep}}$. This means that for any given $\alpha \in \mathfrak{m}_K^{sep}$, there exists a β such that $\pi \cdot \beta = \alpha$ in $M_f(K^{sep})$. It follows that for any given $r \in \mathcal{O}_K$, $rM_f(K^{sep}) = M_f(K^{sep})$ which means that $M_f(K^{sep})$ is divisible $\mathcal{O}_K$-module. Then Λ_f is a divisible torsion $\mathcal{O}_K$-module. Since $\mathcal{O}_K$ is a PID, then we know that Λ_f is isomorphic to $K/\mathcal{O}_K$. Indeed, Λ_f is torsion module then there is no copies of K. Since π is the only prime element in $\mathcal{O}_K$ and $(\mathcal{O}_K)_{(\pi)} \cong \mathcal{O}_K$, then Λ_f is a copy of $K/\mathcal{O}_K$. Moreover, $\Lambda_{f,1}$ is a one dimensional vector space over residue field k (since $\Lambda_{f,1}$ has the same cardinality of residue field k), then $\Lambda_f \cong K/\mathcal{O}_K$. Last claim follows by the fact that $|J| = \dim_k M_f(K^{sep})[\pi] = \dim_k \Lambda_{f,1} = 1$, where J is the index set of copies of $K/\mathcal{O}_K$. □

There is a natural homomorphism $\mathrm{Gal}(K^{sep}/K) \to \mathrm{End}(\Lambda_f)$, this follows from the action of elements in the Galois group on roots of a polynomial. Moreover, since $\Lambda_f \cong K/\mathcal{O}_K$ and $\mathrm{End}(K/\mathcal{O}_K) \cong \mathcal{O}_K$, then it induces a homomorphism $\mathrm{Gal}(K^{sep}/K) \to \mathrm{Aut}(K/\mathcal{O}_K) \cong U_K$.

Theorem 2.2.3 ([2, Chap. 6 Theorem 3])**.** *The natural homomorphism* $\mathrm{Gal}(K^{sep}/K) \to \mathrm{Aut}(\Lambda_f)$ *is an isomorphism.*

Proof. This map is injective by definition; it remains to prove that it is surjective. We have an injection $\mathrm{Gal}(L_{\pi,n}/K) \to U_K/U_K^n$, where $U_K = 1 + \pi^n\mathcal{O}_K$. Let $\alpha \in \Lambda_{f,n}$ be a primitive element, namely those elements such that $[\pi^{n-1}]_f\alpha \neq 0$ or $\pi^{n-1} \cdot \alpha \neq 0$ in $M_f(K^{sep})$. To be precise, α is one of the roots of polynomial $f^{(n)}(X)/f^{(n-1)}(X)$, where $f^{(n)}(X)$ is n-th iterate of $f(X)$. Since $f(X)/X = X^{q-1} + \pi$, then

$$\frac{f^{(n)}(X)}{f^{(n-1)}(X)} = f^{(n-1)}(X)^{q-1} + \pi$$

By the Eisenstein criterion, this is an irreducible polynomial with degree $q^n - q^{n-1}$. Thus, according to the Galois theory, the order of $\mathrm{Gal}(L_{\pi,n}/K)$ is at least $(q-1)q^{n-1}$. On the other hand, U_K/U_K^m has exactly $(q-1)q^{n-1}$ elements. Therefore, $\mathrm{Gal}(L_{\pi,n}/K) \cong U_K/U_K^n$. Eventually, we have that

$$\mathrm{Gal}(K^{sep}/K) \cong \varprojlim \mathrm{Gal}(L_{\pi,n}/K) \cong \varprojlim U_K/U_K^n \cong U_K.$$

□

Corollary 2.2.4 ([2, Chap. 6 Theorem 3])**.** *The element π is a norm from $L_{\pi,n}$.*

Proof. Let $\alpha \in \Lambda_{f,n}$ be a primitive element, then $f^{(n)}(X)/f^{(n-1)}(X)$ is a monic polynomial with constant coefficient π, then the norm $-\alpha$ is π. □

Let K^{un} be the maximal unramified extension of K. Since L_π is a totally ramified extension of K, L_π and K^{un} are linearly disjoint. As a result, the Galois group $\mathrm{Gal}(L_\pi K^{un}/K)$ is a product of $\mathrm{Gal}(L_\pi/K)$ and $\mathrm{Gal}(K^{un}/K)$. Given a uniformizer $\pi \in \mathcal{O}_K$, we define a group homomorphism $r_\pi : K^\times \to \mathrm{Gal}(L_\pi K^{un}/K)$ such that

- $r_\pi(\pi)$ acts as identity on L_π and acts as Frobenius automorphism σ on K^{un}.
- Given a $u \in U_K$, we define $r_\pi(u)$ to act as $[u^{-1}]_f$ on L_π and acts as identity on K^{un}.

Next, one can prove that the field L_π and the homomorphism r_π are independent of the uniformizer π, and that L_π is a maximal abelian extension of K. Moreover, r_π coincides with the local reciprocity map in local class field theory, i.e. for all $a \in \mathcal{O}_K$, $r_\pi(a) = (a, L_\pi K^{un}/K^\times)$. The references of these can be found in [9] and [2].

2.3 Connection with functional equation lemma

In this section, we discuss the connection between the formal group laws obtained from the functional equation lemma and Lubin–Tate formal group laws. Throughout this section we fix a nonarchimedean local field K, let $\mathcal{O}_K$ be its ring of integers, π be a uniformizer, q be the cardinality of the residue field of $\mathcal{O}_K$.

Proposition 2.3.1. *The Lubin-Tate formal group laws F_f associated to $f \in \mathcal{F}_\pi$ are in one-to-one correspondence with formal group laws obtained by the functional equation with parameters $I = (\pi), \sigma = \mathrm{id}, s_1 = \pi^{-1}$ and $\forall i \geq 2, s_i = 0$. The functional equation in the sense of these parameters is*

$$f_g(X) = g(X) + \pi^{-1} f_g(X^q).$$

And this correspondence is given by

$$g(X) \mapsto f_g^{-1}(\pi f_g(X)) \in \mathcal{F}_\pi.$$

Proof. The integrality of the coefficients in $f_g^{-1}(\pi f_g(X))$ follows from the functional equation lemma. Multiplying both sides of the functional equation by π, we have that

$$\pi f_g(X) = \pi g(X) + \pi^{-1}\pi f_g(X).$$

Then $\pi f_g(X) = f_{\pi g}(X)$. According to the second part of the functional equation lemma, we have that

$$f_g^{-1}(\pi f_g(X)) = f_g^{-1}(f_{\pi g}(X)) \in \mathcal{O}_K[\![X]\!].$$

The formal group law obtained by the functional equation lemma is

$$F(X,Y) = f_g^{-1}(f_g(X) + f_g(Y)).$$

By the existence and uniqueness in the theorem 2.1.4, it is enough to check that $f_g^{-1}(\pi f_g(X))$ is a homomorphism for $F(X,Y)$. This can be done as follows:

$$\begin{aligned} f_g^{-1}(\pi f_g(F(X,Y))) &= f_g^{-1}(\pi f_g(f_g^{-1}(f_g(X) + f_g(Y)))) \\ &= f_g^{-1}(\pi f_g(X) + \pi f_g(Y)) \\ &= F(f_g^{-1}(\pi f_g(X)), f_g^{-1}(\pi f_g(Y))). \end{aligned}$$

□

3 Lean and Mathlib

The practice of mathematics is fundamentally rooted in the rigorous verification of proofs. However, as mathematical theories grow increasingly complex, the traditional peer-review process becomes strained. The digitization and formal verification of mathematics offer a modern solution to this problem, allowing mathematicians to verify their results with absolute certainty using software.

This chapter introduces the Lean theorem prover and the mathematics library in the software system called `Mathlib`.

3.1 The Lean Theorem Prover

Lean is a functional programming language and interactive theorem prover initially developed by Leonardo de Moura at Microsoft Research in 2013 [4]. The current version, Lean 4, is a complete reimplementation that serves simultaneously as a proof assistant and a general-purpose, practically efficient programming language.

At its logical core, Lean is based on a variant of dependent type theory known as the Calculus of Inductive Constructions (CIC). In this framework, propositions are represented as types, and proofs are represented as terms of those types—a correspondence known as the Curry-Howard isomorphism.

To prove a theorem in Lean is to construct a term of the corresponding type. Lean's kernel, a small and highly trusted piece of code, checks that this term is well-typed. If the kernel accepts the term, the proof is logically sound. This minimizes the trusted code base; users do not need to trust the complex tactics or automation used to generate the proof, only the kernel that verifies the final proof term.

3.2 Dependent Type Theory

Traditional set theory (such as Zermelo-Fraenkel with Choice, ZFC) has historically served as the foundation of mathematics. However, dependent type theory provides a more natural environment for machine verification.

In Lean's type theory, every expression has a specific type. The defining feature of *dependent* type theory is that types can depend on terms. For example, the type of a vector of length n, denoted `Vector a n`, depends on the natural number n. This allows for highly expressive typing where the properties of mathematical objects are baked directly into their definitions.

Moreover, even types themselves have types. The type of propositions `Prop` is also of type `Type`. As to not run into a paradox called Girard's paradox there is a hierarchy of types in [3]. The type of `Type` is `Type 1`, the type of `Type 1` is `Type 2` and so on. These are call type-universe. We can also say `R` is a type in an arbitrary univer and we denote it as `R : Type*`.

3.3 Inductive type and Quotient type

3.3.1 Inductive type

In the Lean proving system, the vast majority of mathematical objects, data structures, and logical propositions are defined as *inductive types*.

An inductive type is defined by specifying its universe level and a comprehensive list of its *constructors*. Mathematically, inductive types are "freely generated" by these constructors. Here are several examples in Mathlib

```
inductive Nat where
  | zero : Nat
  | succ (n : Nat) : Nat

inductive List (α : Type u) where
  | nil : List α
  | cons (head : α) (tail : List α) : List α

inductive Option (α : Type u) where
  /-- No value. -/
  | none : Option α
  /-- Some value of type `α`. -/
  | some (val : α) : Option α
```

In the inductive construction of `List`, `nil` means the empty list, usually written `[]`. And the inductive construction of `List` is that: the list whose first element is `head`, where `tail` is the rest of the list. Usually written `head :: tail`.

Because inductive types are freely generated, Lean automatically enforces two strict rules, often referred to as the "no confusion" properties [5]:

1. Injectivity of Constructors: If $\texttt{succ}(a) = \texttt{succ}(b)$, then $a = b$.
2. Disjointness of Constructors: Different constructors can never produce the same value. For example, $\texttt{zero} \neq \texttt{succ}(a)$ for any a.

These properties are what allow functions to be defined via pattern matching and recursive evaluation. Is each type in Lean an inductive type? The answer is almost but not. Every type in Lean is either an inductive type, or defined in terms of universes (like the type `Type 2`), functions and quotient types.

3.3.2 Quotient types

In abstract algebra and topology, it is frequently necessary to construct a new space by "gluing" elements of a base space together according to an equivalence relation $\sim$.

If one were to attempt to define a quotient type as a standard inductive type, they would face a contradiction. One might try to provide a constructor $\texttt{mk} : \alpha \to \text{Quotient}$, but because inductive constructors are strictly injective, Lean would enforce that if $a \neq b$, then $\texttt{mk}(a) \neq \texttt{mk}(b)$. This entirely defeats the purpose of a quotient, where we explicitly want $\texttt{mk}(a) = \texttt{mk}(b)$ whenever $a \sim b$.

To resolve this, Lean does not treat quotients as inductive types. Instead, it extends its core type theory with a set of primitive axioms specifically designed for quotienting [1].

Lean introduces four foundational constants to handle quotients for any type α and relation $r : \alpha \to \alpha \to \texttt{Prop}$:

1. `Quot` is the built-in quotient type.
2. `Quot.mk` places elements of the underlying type α into the quotient.
3. `Quot.lift` allows the definition of functions from the quotient to some other type.
4. `Quot.ind` is used to write proofs about quotients by assuming that all elements are constructed with `Quot.mk`.

```
axiom Quot.sound : ∀ {α : Sort u} {r : α → α → Prop} {a b : α},
    r a b → Quot.mk r a = Quot.mk r b
```

By utilizing `Quot.sound`, Lean intentionally breaks the injectivity rule that governs inductive types, allowing for true mathematical equivalence classes. Functions out of quotient types are then defined using `Quot.lift`, which requires the user to provide a proof that the function respects the underlying equivalence relation (i.e., the function is well-defined).

3.4 Structures, Classes, and Type Class Inference

To effectively formalize mathematics, a system must be able to organize abstract concepts—such as groups, rings, and topological spaces—in a modular and extensible way. Lean achieves this using structures and type classes.

3.4.1 Structures and Classes

In Lean, a `structure` is an inductive type with a single constructor, akin to a record in traditional programming languages. It allows users to bundle data (such as mathematical operations) and propositions (such as axioms) into a single cohesive object. For example, a group can be defined as a structure containing a carrier type, a multiplication operation, an identity element, an inverse operation, and the explicit proofs that these operations satisfy the standard group axioms.

A `class` is simply a `structure` that has been registered with Lean's type class resolution system. In mathematical formalization, we almost exclusively use classes to define algebraic and topological properties. By doing so, we define generic interfaces that many different specific types (e.g., the integers, polynomials, or matrices) can instantiate.

3.4.2 Type Class Inference

Type class inference is the automated mechanism by which Lean synthesizes missing mathematical context. When a theorem or definition requires an instance of a class—denoted by square brackets in Lean's syntax, such as `[Ring R]`—Lean searches its environment to automatically construct and inject the required instance.

This system is the absolute backbone of mathematical notation and generic reasoning in Lean. For instance, the addition symbol `+` is defined via a fundamental type class called `Add`. When Lean encounters the expression `a + b` for variables `a b : R`, it implicitly triggers type class inference to find an instance of `Add R`. If `R` is known to be a ring via a local assumption, Lean seamlessly infers the correct addition operation without the user needing to manually specify it. This mechanism allows formalized mathematics to remain readable and closely resemble traditional pen-and-paper mathematics.

3.4.3 The Algebraic Hierarchy

In mathlib, mathematical structures are organized into a massive, heavily intertwined algebraic hierarchy. This hierarchy is built using class extension.

```
class Semiring (α : Type u) extends NonUnitalSemiring α, NonAssocSemiring α,
    MonoidWithZero α

class Ring (R : Type u) extends Semiring R, AddCommGroup R, AddGroupWithOne R

class CommRing (α : Type u) extends Ring α, CommMonoid α
```

For instance, the `CommRing` (commutative ring) class extends the `Ring` class, which in turn extends both `Semiring`, `AddCommGroup` and `AddGroupWithOne`, cascading all the way down to basic operational classes like `Mul` and `Add`.

Because of this hierarchy, a theorem proven for a `Monoid` is automatically applicable to a `Group`, a `Ring`, or a `Field`. Type class inference handles this "forgetful inheritance" automatically. When a theorem requires a `Monoid` but is given a `Ring`, Lean's type class synthesis systematically the hierarchy to extract the underlying `Monoid` structure.

Designing and maintaining this hierarchy is one of the most significant software engineering achievements within the mathlib project. It ensures maximal theorem reuse, preventing mathematicians from having to prove the same fundamental lemmas multiple times across different domains of algebra.

4 Lean formalization of Formal Group Laws

In this chapter we discuss our formalization of formal group laws.

4.1 Definitions

We choose to formalize the definition of formal group laws in Lean using the definition in 1.1.1, since this is equivalent to the definitions in other references and easier to check in practice.

The following is our definition of formal group laws in Lean:

```
@[ext]
structure FormalGroup (R : Type*) [CommRing R] where
  toFun : MvPowerSeries (Fin 2) R
    /- Constant coefficient of `F` is zero. -/
    zero_constantCoeff : constantCoeff toFun = 0
    /- Coefficient of `X` and `Y` is 1. -/
    lin_coeff_X : coeff (single 0 1) toFun = 1
    lin_coeff_Y : coeff (single 1 1) toFun = 1
    /- Associativity condition: `F (F (X, Y), Z) = F (X, F (Y, Z))`.-/
    assoc : subst ![subst ![Y₀, Y₁] toFun, Y₂] toFun = subst ![Y₀, subst ![Y₁, Y₂]
    toFun] toFun (S := R)
```

In Mathlib, `MvPowerSeries` σ `R` is the ring of multivariate power series with indeterminates indexed by σ. Also, `Fin n` is a type with exactly n elements. To be specific, an element of `Fin n` is an `i : ℕ` together with a proof `i < n`. Moreover, `MvPowerSeries.X (i :` σ`)` is the indeterminate indexed by `i`, and in the namespace `MvPowerSeries` we can abbreviate `MvPowerSeries.X` as `X`. Therefore, `X 0`, `X 1`, and `X 2` are the three indeterminates in `MvPowerSeries (Fin 3) R`. The notation `Y₀, Y₁, Y₂` is an abbreviation for `X 0, X 1, X 2` in `MvPowerSeries (Fin 3) R`. The expression `![a, b]` in the associativity condition is a map from `Fin 2` to α, where `a b :` α. `subst` is the function used to express substitution of (multivariate) power series. For instance, $f(g(X_I))$ is expressed as `f.subst g` and $F(f(X_I), g(X_I))$ is expressed as `F.subst ![f, g]`.

For the definition of commutative formal group laws, the only additional condition is commutativity.

```
@[ext]
structure CommFormalGroup (R : Type*) [CommRing R] extends FormalGroup R where
/- Commutativity condition: `F(X,Y) = F(Y,X)`.-/
comm : toFun = subst ![X₁, X₀] toFun
```

Here, `X₀, X₁` is an abbreviation for `X 0, X 1` in `MvPowerSeries (Fin 2) R`.

Mathematically, a commutative formal group is a formal group. In Lean, this statement is expressed via coercions. An instance is as follows:

```
/-- A commutative formal group law is a formal group law.-/
instance : Coe (CommFormalGroup R) (FormalGroup R) where
coe := CommFormalGroup.toFormalGroup
```

Given a formal group law, it defines a group structure on `MvPowerSeries σ R`. To be precise, let `(F : FormalGroup R) (f₀ f₁ : MvPowerSeries σ R)`. The sum of `f₀` and `f₁` is defined as `subst ![f₀, f₁] F.toFun`. In Lean, a simpler notation can be written as

```
abbrev add (F : FormalGroup R) (f g : MvPowerSeries σ R) : MvPowerSeries σ R :=
    subst ![f, g] F.toFun
```

In the environment of `namespace FormalGroup`, `+[F]` is the addition with respect to `F : FormalGroup R`. To be specific, given two multivariate power series $f, g \in R[\![X_I]\!]$ where I is an index set, and the constant coefficients of f, g are zero, `f +[R] g` means $F(f, g)$.

With this notation, we can rewrite the associativity condition and the commutativity condition in the `namespace FormalGroup` as follows:

```
/-- The addition under the sense of formal group `F` is associative. -/
theorem add_assoc {F : FormalGroup R} {Z₀ Z₁ Z₂ : MvPowerSeries σ R} (hZ₀ :
    constantCoeff Z₀ = 0) (hZ₁ : constantCoeff Z₁ = 0) (hZ₂ : constantCoeff Z₂ = 0):
Z₀ +[F] Z₁ +[F] Z₂ = Z₀ +[F] (Z₁ +[F] Z₂) := ...

theorem add_comm {F : FormalGroup R} (hF : F.comm) {Z₀ Z₁ : MvPowerSeries σ R}
(hZ₀ : constantCoeff Z₀ = 0) (hZ₁ : constantCoeff Z₁ = 0):
  Z₀ +[F] Z₁ = Z₁ +[F] Z₀ := ...
```

To make sense of the fact that a formal group law F defines a formal addition, we should verify that there is a unit for this addition, which means $F(X, 0) = X$ and $F(0, Y) = Y$. Furthermore, we should prove that there exists a unique additive inverse element.

The existence of an additive unit in the sense of F is formalized by the following two theorems.

```
theorem zero_left : subst ![0, PowerSeries.X] F.toFun = PowerSeries.X (R := R) := ...

theorem zero_right : subst ![PowerSeries.X, 0] F.toFun = PowerSeries.X (R := R) := ...
```

Using the same notation for formal group addition, we can generalize the above theorems to the following two theorems.

```
theorem zero_add (h : constantCoeff f = 0) : 0 +[F] f = f :=

theorem add_zero (h : constantCoeff f = 0) : f +[F] 0 = f :=
```

Here, the name `zero_add` means that adding zero to a multivariate power series `f` gives `f` itself, which is a standard naming convention in Mathlib. To distinguish this theorem from other theorems with the same name, we place it in the `namespace FormalGroup`.

4.1.1 Examples of formal groups

The basic examples of formal group laws are the additive formal group law $\mathbf{G}_a$ and the multiplicative formal group law $\mathbf{G}_m$. We formalize them via our definition of `FormalGroup R`.

```
def Gₐ : CommFormalGroup R where
  toFun := X₀ + X₁
  ...

def Gₘ : CommFormalGroup R where
  toFun := X₀ + X₁ + X₀ * X₁
  ...
```

Now we discuss the construction of the additive inverse of a formal group law.

```
abbrev addInv_aux (F : FormalGroup R) : ℕ → R
  | 0 => 0
  | 1 => -1
  | n + 1 => - (coeff (n + 1) (F.toFun.subst
    ![X, (∑ i : Fin (n + 1), C (addInv_aux F i.1) * X ^ i.1)]))

def addInv_X : PowerSeries R := .mk (addInv_aux F ·)

def addInv (φ : MvPowerSeries σ R) : MvPowerSeries σ R := subst φ (addInv_X F)
```

The API `addInv_X` is the additive inverse of `X` in the sense of formal group law `F`. For arbitrary multivariate power series, `addInv` φ is the additive inverse of φ. Via direct calculation and induction, we have the theorem

```
theorem subst_addInv_eq_zero : F.toFun.subst ![X, (addInv_X F)] = 0 := by
```

4.1.2 Functions Between Formal Groups

According to 1.1.11, given two commutative rings R, R' and a ring homomorphism $\phi : R \to R'$, and a formal group law F over R, then $\phi_* F$ is a formal group law over R'. The construction of the pushforward on the coefficient ring in Lean is:

```
def map {R : Type*} [CommRing R] {R' : Type*} [CommRing R']
    (f : R →+* R') (F : FormalGroup R): FormalGroup R' where
  toFun := F.toFun.map f
  ...
```

Let R be a commutative ring and T be a subring of R. Given a (multivariate) power series f over R with the assumption that all coefficients of f are in the subring T, f can be considered as a (multivariate) power series over T. We define this function as `MvPowerSeries.toSubring`, following the APIs for polynomials as inspiration.

```
variable [Ring R] (p : MvPowerSeries σ R) (T : Subring R) (hp : ∀ n, p.coeff n ∈ T)

def MvPowerSeries.toSubring : MvPowerSeries σ T := fun n => ⟨p.coeff n, hp n⟩
```

According to the definition, their coefficients should be the same after coercion. The `@[simp]` tag in the following theorem means that this theorem can be applied automatically when we use the tactic `simp` to simplify an expression.

```
@[simp]
theorem coeff_toSubring {n : σ →₀ ℕ} : (p.toSubring T hp).coeff n = p.coeff n := rfl
```

Given a formal group law F over a ring R with the assumption that all coefficients of F are in the subring T, then F is a formal group law over T.

```
def FormalGroup.toSubring (F : FormalGroup R) (hF : ∀ n, F.toFun n ∈ T) :
    FormalGroup T where
  toFun := F.toFun.toSubring _ hF
  ...
```

Similarly, we can also define `CommFormalGroup.toSubring`. Since we have a coercion from `CommFormalGroup R` to `FormalGroup R`, we only need to check the commutativity condition.

```
lemma CommFormalGroup.toSubring_comm (F : CommFormalGroup R) (hF : ∀ n, F.toFun n ∈
    T) :
    F.toFun.toSubring _ hF = (F.toFun.toSubring _ hF).subst ![X₁, X₀] := ...
```

```
def CommFormalGroup.toSubring (F : CommFormalGroup R) (hF : ∀ n, F.toFun n ∈ T) :
    CommFormalGroup T where
  toFun := F.toFun.toSubring _ hF
  zero_constantCoeff := ((F : FormalGroup R).toSubring _ hF).zero_constantCoeff
  lin_coeff_X := ((F : FormalGroup R).toSubring _ hF).lin_coeff_X
  lin_coeff_Y := ((F : FormalGroup R).toSubring _ hF).lin_coeff_Y
  assoc := ((F : FormalGroup R).toSubring _ hF).assoc
  comm := F.toSubring_comm _ hF
```

4.1.3 Homomorphism and Isomorphism

First, we need the API for `PowerSeries.toMvPowerSeries` which is inspired by a similar API for polynomials.

```
noncomputable
def Polynomial.toMvPolynomial (i : σ) : R[X] →ₐ[R] MvPolynomial σ R :=
  aeval (MvPolynomial.X i)
```

Given a power series $p(X) \in R[\![X]\!]$ and an index $i \in I$ where I is an index set, this is a function that maps $p(X)$ to $p(X_i) \in R[\![X_I]\!]$.

```
def PowerSeries.toMvPowerSeries (f : PowerSeries R) (i : σ) : MvPowerSeries σ R :=
  f.subst (MvPowerSeries.X i)
```

For the definition of homomorphism 1.1.5 between two formal group laws, it is defined as `structure` in Lean:

```
@[ext]
structure FormalGroupHom  (G₁ G₂ : FormalGroup R) where
  toFun : PowerSeries R
  zero_constantCoeff : toFun.constantCoeff = 0
  hom : toFun.subst (G₁.toFun) = G₂.toFun.subst (R := R) toFun.toMvPowerSeries
```

Since two formal group homomorphisms are the same if and only if their underlying functions are the same, we add a tag `@[ext]` to the structure. When `@[ext]` is applied to a structure, it generates an `.ext` theorem and applies the `@[ext]` attribute to it. The result is an `.ext` and an `.ext_iff` theorem with the `.ext` theorem registered for the `ext` tactic.

An isomorphism between two formal groups is defined as two formal group homomorphisms that are inverse to each other.

```
@[ext]
structure FormalGroupIso (G₁ G₂ : FormalGroup R) where
  toHom : FormalGroupHom G₁ G₂
  invHom : FormalGroupHom G₂ G₁
  left_inv : (PowerSeries.subst toHom.toFun) ∘ (PowerSeries.subst invHom.toFun) = id
  right_inv : (PowerSeries.subst invHom.toFun) ∘ (PowerSeries.subst toHom.toFun) = id
```

Here we also add a tag `@[ext]` for the `structure` `FormalGroupIso` and we automatically obtain a theorem

```
variable {G₁ G₂ : FormalGroup R} {x y : FormalGroupIso G₁ G₂}

theorem FormalGroupIso.ext_iff :
    x = y ↔ x.toHom = y.toHom ∧ x.invHom = y.invHom := ...
```

To show that two formal group isomorphisms are the same, it is enough to show that their `toHom` is the same. Indeed, if we have `x.toHom = y.toHom`, then `x.invHom = y.invHom` is also true since `invHom` is the substitution inverse of `toHom` and the substitution inverse is unique. And the following theorem is more convenient to use in practice.

```
theorem FormalGroupIso.ext_iff' : x = y ↔ x.toHom = y.toHom := ...
```

Let $F(X,Y), G(X,Y)$ be formal group laws over a ring R. A homomorphism of formal group laws $\alpha(X)$ from $F(X,Y)$ to $G(X,Y)$ is an isomorphism if and only if the coefficient of X in $\alpha(X)$ is invertible in R.

```
theorem isIso_of_isUnit_coeff_one {G₁ G₂ : FormalGroup R} (α : FormalGroupHom G₁ G₂)
    (h : IsUnit (α.toFun.coeff 1)) :
    ∃ (α₁ : FormalGroupIso G₁ G₂), α₁.toHom = α := by

theorem isUnit_coeff_one_of_isIso {G₁ G₂ : FormalGroup R} (α : FormalGroupIso G₁ G₂) :
    IsUnit (α.toHom.toFun.coeff 1) := by
```

There are two ways to express an element `a` of a ring `R` to be invertible. The first one is `IsUnit a`, while the second one is `Invertible a`. I briefly explain their difference.

The type `IsUnit a` is defined as a `Prop` (a proposition). Under the hood, it asserts that there exists some unit u in the group of units $R^\times$ such that u, coerced to R, equals a. It merely states the fact that an inverse exists. It tells Lean's typechecker, "I promise an inverse exists somewhere.". A limitation is that, since it is a `Prop`, it is subject to proof irrelevance. Lean erases propositions during computation. One cannot easily extract the inverse from `IsUnit a` to use in a computable function without invoking the Axiom of Choice (which makes the resulting definition `noncomputable`).

Conversely, `Invertible a` is a typeclass (residing in `Type`) that explicitly packages the element a with its concrete inverse. It conceptually contains the inverse element itself, alongside the proofs that multiplying them together yields 1 in the ring. It provides explicit, computable access to the inverse. Because it is a typeclass, Lean's typeclass synthesis system can automatically find and apply inverses. Also, having an `[Invertible a]` instance in the local context allows the use of the special notation to write the inverse of a directly in algebraic expressions.

```
theorem isUnit_of_invertible [Monoid α] (a : α) [Invertible a] : IsUnit a :=
```

The theorem `isUnit_of_invertible` in Mathlib gives one direction of the equivalence between these two APIs.

In the proof of these two theorems, we need the result that, given a power series $f(X)$ over a ring R, if the coefficient of X in $f(X)$ is invertible and the constant coefficient is zero, then there is a power series $g(X)$ over R, such that $f(g(X)) = g(f(X)) = X$.

```
variable (P : PowerSeries R) (hP : P.constantCoeff = 0) [Invertible (coeff 1 P)]

noncomputable
def substInvFun : ℕ → R
  | 0 => 0
  | 1 => 1 / (coeff _ 1 P)
  | n + 1 => - 1 / (coeff 1 P) *
      (coeff (n + 1) (P.subst (∑ i : Fin (n + 1), C (substInvFun i.1) * X ^ i.1)))
```

According to the construction and computation, we have the following theorem

```
lemma substInv_subst : P.substInv.subst P = X :=
```

Also, we can prove that this substitution inverse is unique.

4.2 Functional Equation Lemma

In this section, we discuss the formalization of the functional equation lemma. To begin with, we formalize the data involved in the functional equation lemma as follows.

```
variable {K : Type*} [CommRing K] {R : Subring K} {I : Ideal R}
  {p t q : ℕ} [hp : Fact (Nat.Prime p)] (ht : t ≠ 0) (hq : q = p ^ t)
  (σ : K →+* K) (s : ℕ → K) {g : PowerSeries R}
```

Let K be a commutative ring, R a subring of K, and I an ideal of R. Let p, q, t be natural numbers, where p is a prime number and q can be written as $q = p^t$ with nonzero t. Let σ be a ring endomorphism of K, and let s be a sequence of elements in K.

According to equation 1.2.2.1, we can inductively define a function `RecurFunAux` using equation 1.2.2.1 and then obtain a power series using the API `PowerSeries.mk`. We then prove that it satisfies the functional equation 1.2.1.1, namely

$$f_g(X) = g(X) + \sum_{i=1}^{\infty} s_i \sigma_*^i f_g(X^{q^i}).$$

```
def RecurFunAux (hg : g.constantCoeff = 0) : ℕ → K
  | 0 => 0
  | k + 1 =>
    g.coeff (k + 1) + ∑ j ∈ (Icc 1 (multiplicity q (k + 1))).attach,
      have : ((k + 1) / (q ^ (j : ℕ))) < k + 1 := ...
      (s j) * σ^[j] (RecurFunAux hg ((k + 1) / (q ^ (j : ℕ))))

def RecurFun := PowerSeries.mk (RecurFunAux ht hq σ s hg)
```

In order to inductively define `RecurFunAux`, we prove that for all `j ∈ Icc 1 (multiplicity q (k + 1))`, `((k + 1) / (q ^ (j : ℕ))) < k + 1` in the motive of `k + 1`. Therefore, we need to use `Finset.attach`, which enables us to use the property `j ∈ Icc 1 (multiplicity q (k + 1))`. To be precise, let `s` be a finite set; `Finset.attach s` takes the elements of `s` and forms a new set of elements of the subtype `{x // x ∈ s}`. With the property `j ∈ Icc 1 (multiplicity q (k + 1))`, we can prove that `((k + 1) / (q ^ (j : ℕ))) < k + 1`. Otherwise, `(RecurFunAux hg ((k + 1) / (q ^ (j : ℕ))))` is not valid.

To state the formalization of the functional equation, we need APIs related to `expand`, which is an algebraic homomorphism that expands a (multivariate) power series by a nonzero factor p. We discuss these APIs first.

```
variable {σ τ R S : Type*}  [CommRing R] [CommRing S] (p : ℕ) (hp : p ≠ 0)

/-- Expand the power series by a factor of p, so `∑ aₙ x^n` becomes `∑ aₙ x^(np)`. -/
noncomputable def expand : MvPowerSeries σ R →ₐ[R] MvPowerSeries σ R :=
  substAlgHom (HasSubst.X_pow hp)
```

We formalize functional equation 1.2.1.1 as follows.

```
theorem FunEq_of_RecurFun [TopologicalSpace K] [T2Space K] (hs₀ : s 0 = 0) :
    let f := RecurFun ht hq σ s hg
    f = g.map R.subtype + ∑' (i : ℕ), s i · (f.expand (q ^ i) (q_pow_neZero hq)).map
    (σ^i) := ...
```

The additional instance `[TopologicalSpace K]` provides a topological structure on the ring of power series over K, and `[T2Space K]` ensures that the infinite sum in the equation has a

unique limit whenever it converges. Let R be a subring of K. Then `R.subtype` is a canonical ring homomorphism from R to K. `PowerSeries.expand p hp f` expands a power series `f` by a nonzero factor p.

The theorem `coeff_one_RecurFun` states that the 1st coefficient of g equals the 1st coefficient of $f_g(X)$. With the assumption `(hg_unit : IsUnit (g.coeff 1))`, we obtain `IsUnit ((RecurFun ht hq σ s hg).coeff 1)`. Therefore, there is a substitution inverse of $f_g(X)$, denoted by $f_g^{-1}(X)$.

```
lemma coeff_one_RecurFun: (RecurFun ht hq σ s hg).coeff 1 = g.coeff 1 := ...

lemma coeff_RecurFun_unit (hg_unit : IsUnit (g.coeff 1)) :
    IsUnit ((RecurFun ht hq σ s hg).coeff 1) := ...

def inv_RecurFun := PowerSeries.subst_inv _ (coeff_RecurFun_unit ht hq σ s hg hg_unit)
  (constantCoeff_RecurFun ..)

def inv_add_RecurFun :=
    (inv_RecurFun ht hq σ s hg hg_unit).subst
      ((RecurFun ht hq σ s hg).toMvPowerSeries (0 : Fin 2) +
        (RecurFun ht hq σ s hg).toMvPowerSeries 1)
```

The API `inv_add_RecurFun` is the multivariate power series $F_g(X,Y) = f_g^{-1}(f_g(X) + f_g(Y))$. Also, `(RecurFun ht hq σ s hg).toMvPowerSeries (0 : Fin 2)` represents the multivariate power series $f_g(X)$, where X is the first indeterminate variable in a bivariate power series. Similarly, `(RecurFun ht hq σ s hg).toMvPowerSeries 1` represents $f_g(Y)$.

Recall that the first part of the functional equation lemma 4 says that $F_g(X,Y)$ has coefficients in R. Moreover, this is a commutative formal group law by the definition of $F_g(X,Y)$. The first part of the functional equation lemma is formalized as follows.

```
theorem coeff_inv_add_mem_Subring [UniformSpace K] [T2Space K] [DiscreteUniformity K]
    (hs₀ : s 0 = 0) : ∀ n, (inv_add_RecurFun ht hq σ s hg hg_unit).coeff n ∈ R := ...
```

Next, we first prove that $F_g(X,Y)$ is a formal group law over K. The associativity condition follows from

$$F_g(F_g(X,Y),Z) = f_g^{-1}(f_g(X) + f_g(Y) + f_g(Z)) = F_g(X, F_g(Y,Z))$$

Then we use the API `FormalGroup.toSubring` and the theorem `coeff_inv_add_mem_Subring` to obtain a formal group law `CommFormalGroup.InvAdd_RecurFun` over R.

```
def CommFormalGroup.InvAdd_RecurFun_Aux : CommFormalGroup K where
  toFun := inv_add_RecurFun ht hq σ s hg hg_unit
  ...

def CommFormalGroup.InvAdd_RecurFun [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs₀ : s 0 = 0) : CommFormalGroup R :=
  (CommFormalGroup.InvAdd_RecurFun_Aux ht hq σ s hg hg_unit).toSubring _
    (coeff_inv_add_mem_Subring ...)
```

Recall that the second part of the functional equation lemma 4 states that, for any power series g, g' over R with the assumption that the constant coefficients of g, g' are zero, then $f_{g'}^{-1}(f_g(X))$ has coefficients in R. The second part of the functional equation lemma is formalized as follows. In the lemma below, `f_g'` is the power series $f_{g'}(X)$ and `G` is $f_{g'}^{-1}(f_g(X))$.

```
lemma coeff_inv_RecurFun_g'_mem_Subring [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs0 : s 0 = 0) :
    let f_g' := (RecurFun ht hq σ s hg')
    let G := (inv_RecurFun ht hq σ s hg hg_unit).subst f_g'
    ∀ n, PowerSeries.coeff n G ∈ R
```

The third part of the functional equation lemma states that there is a power series $\hat{h}$ over R such that $f_g(h(X)) = f_{\hat{h}}(X)$. According to the functional equation 1.2.1.1, this is equivalent to showing that the power series $f_g(h(X)) - \sum_{n=1}^{\infty} s_n \cdot \sigma^n f_g(h(X^{q^n}))$ is a power series over R. This is exactly the desired power series $\hat{h}$.

```
lemma coeff_inv_RecurFun_g'_mem_Subring' [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs₀ : s 0 = 0) {h : PowerSeries R}
    (h₀ : h.constantCoeff = 0):
    let f := (RecurFun ht hq σ s hg)
    let f₁ : PowerSeries K := f.subst (h.map R.subtype)
    ∀ n, (f₁ - ∑' i, (s i) · (f₁.expand (q ^ i) (q_pow_neZero hq)).map (σ ^ i)).coeff
    n ∈ R :=

theorem congr_equiv [UniformSpace K] [T2Space K] [DiscreteUniformity K]
    (hs₀ : s 0 = 0) {α : PowerSeries R} {β : PowerSeries K}
    (hα : α.constantCoeff = 0) (hβ : β.constantCoeff = 0)
    {r : ℕ} (hr : 1 ≤ r) :
    let f := RecurFun ht hq σ s hg
    (∀ n, α.coeff n - β.coeff n ∈ R.subtype '' ↑(I ^ r)) ↔
      (∀ n, PowerSeries.coeff n (f.subst (α.map R.subtype)) -
        PowerSeries.coeff n (f.subst β) ∈ R.subtype '' ↑(I ^ r)) :=
```

I briefly explain some details in Part Four of the functional equation lemma. To begin with, `α.coeff n - β.coeff n` is an element in K, since α is a power series over R and β is a power series over K. Therefore, we need to formalize it as an element in a set of K rather than in an ideal of R. Secondly, `R.subtype '' ↑(I ^ r)` is a set of elements in K that is the image of I^r under the natural algebraic map $R \to K$. Moreover, the symbol ↑ is necessary since there is a coercion from `Ideal R` to `Set R`.

4.3 Universal formal group laws

In this section, we discuss some formalization related to universal formal group laws. We fix some environmental variables.

```
variable {R : Type u} [CommRing R] (U : FormalGroup R)
```

To begin with, we define a functor from the category of commutative rings to **Type**, which maps a commutative ring R to formal group laws over R.

```
def FGL : CommRingCat.{u} ⥤ Type u where
  obj R := FormalGroup R
  map f := FormalGroup.map f.hom
  map_id R := by ext F : 1; simp [FormalGroup.map]
  map_comp f g := by ext F : 1; simp [FormalGroup.map]
```

The symbol ⥤ is notation for a functor between two categories. A functor in Lean requires four constructors: `obj`, `map`, `map_id`, `map_comp`. Given an object `R` of type `CommRingCat`, `obj R` means all formal groups over the ring `R`. Moreover, given a morphism in `CommRingCat`, namely `f : X ⟶ Y`, `map f` is the pushforward map defined before.

Recall that a one-dimensional formal group law $F(X,Y)$ over L is called a universal formal group law if and only if, for any formal group law $G(X,Y)$ over R, there is a unique ring homomorphism $\phi : L \to R$ such that $\phi_* F(X,Y) = G(X,Y)$.

```
def IsUniversal : Prop :=
    ∀ {T : Type u} [CommRing T], ∀ F, ∃! (φ : R →+* T), U.map φ = F
```

`IsUniversal` is formalized as a definition of type `Prop` for a formal group law. The `{R : Type u}` and `{T : Type u}` mean that we make these two variables implicit. The reason that we can make them implicit is that `R` and `T` can be inferred from `U` and `F`, respectively. The definition is in the namespace `FormalGroup`, so we use `U.IsUniversal` to express that U is a universal formal group law.

Under the assumption `U.IsUniversal`, we have the following theorem:

```
theorem FGL_representable (h : U.IsUniversal) : FGL.IsCorepresentedBy U (X := of R) :=
    ...
```

First, we briefly explain the API `IsCorepresentedBy`.

```
@[mk_iff]
structure _root_.CategoryTheory.Functor.IsCorepresentedBy
    (F : C ⥤ Type w) {X : C} (x : F.obj X) : Prop where
  map_bijective {Y : C} : Function.Bijective (fun f : X ⟶ Y ↦ F.map f x)
```

A presheaf `F` is represented by `X` with universal element `x : F.obj X` if the natural transformation `F ⟶ yoneda.obj X` induced by `x` is an isomorphism. For better universe generality, we state this manually: for every `Y`, the induced map `(X ⟶ Y) → F.obj Y` is bijective. Adding the tag `@[mk_iff]` automatically generates a theorem as follows:

```
CategoryTheory.Functor.isCorepresentedBy_iff.{u, v, w} {C : Type u}
    [Category.{v, u} C] (F : C ⥤ Type w) {X : C} (x : F.obj X) :
    F.IsCorepresentedBy x ↔ ∀ {Y : C}, Function.Bijective fun f ↦ F.map f x
```

Therefore, this theorem says that if U is a universal formal group law over R, then for any commutative ring S, there is a bijection between $\mathcal{H}om_{Ring}(R,S)$ and formal group laws over S.

4.4 Commutative Lemma

In this section, we discuss the formalization of some commutativity results from the previous chapter. Fix a formal group law $F(X,Y)$ over R throughout this section. First, we fix some environmental variables.

```
variable {R : Type*} [CommRing R] (F : FormalGroup R) [Nontrivial R] {σ : Type*}
```

According to the definition of commutativity for formal group laws, we have that $F(X,Y)$ is commutative if and only if the coefficients of X^iY^j and X^jY^i coincide.

```
theorem comm_iff_coeff_symm :
  F.comm ↔ ∀ (i j : ℕ), coeff (single 0 i + single 1 j) F.toFun
    = coeff (single 0 j + single 1 i) F.toFun := by
```

A precommutator of a formal group law `F` is a multivariate power series `X +[F] Y +[F] (-X)`. Typically, a commutator of a group law is defined to be $xyx^{-1}y^{-1}$, where x, y are elements of a group G, and a group is commutative if and only if its commutator is zero. A formal group law $F(X,Y)$ is commutative if and only if the precommutator is equal to `Y`.

```
def preCommutator (f g : MvPowerSeries σ R) := f +[F] g +[F] (addInv F f)

def commutator : MvPowerSeries (Fin 2) R :=
    X₀ +[F] X₁ +[F] (addInv F X₀) +[F] (addInv F X₁)

lemma preCommutator_ne_of_nonComm (h : ¬ F.comm) :
    preCommutator F X₀ X₁ ≠ X₁ := by

lemma comm_iff_commutator_eq_zero : F.comm ↔ commutator F = 0 := by
```

Now we can formalize Lemma 1.4.2 and Lemma 1.4.3.

```
theorem exists_nonzero_hom_to_Ga_or_Gm_of_not_comm (h : ¬ F.comm) :
    (∃ (α : FormalGroupHom F (Gₐ (R := R))), α.toFun ≠ 0) ∨
    (∃ (α : FormalGroupHom F (Gₘ (R := R))), α.toFun ≠ 0) :=

theorem comm_of_exists_nonzero_hom_to_comm (F' : FormalGroup R) [IsDomain R]
    (α : FormalGroupHom F F') (ha : α.toFun ≠ 0) :
    F'.comm → F.comm := by
```

The instance `IsDomain R` means that `R` is an integral domain. Here $\mathtt{G}_a$ and $\mathtt{G}_m$ are abbreviations for the additive and multiplicative formal group laws. The notation `(R := R)` means that the coefficient ring of the formal group laws is `R`.

Therefore, we can deduce that any formal group law over a domain R is commutative with a neat proof.

```
theorem comm_of_isDomain [IsDomain R] : F.comm := by
  by_contra hc
  obtain ⟨α, ha⟩| ⟨α, ha⟩ := exists_nonzero_hom_to_Ga_or_Gm_of_not_comm F hc
  · exact hc ((comm_of_exists_nonzero_hom_to_comm _ _ α ha) Gₐ.comm)
  · exact hc ((comm_of_exists_nonzero_hom_to_comm _ _ α ha) Gₘ.comm)
```

4.5 Lubin-Tate Theory

In this section, we discuss some formalization of Lubin–Tate formal group laws. Let K be a nonarchimedean local field, let $\mathcal{O}_K$ be its ring of integers, and let π be a uniformizer. Since K is a nonarchimedean local field, the residue field of $\mathcal{O}_K$ is finite.

```
variable {K : Type u} [Field K] [ValuativeRel K] [TopologicalSpace K]
  [IsNonarchimedeanLocalField K] (π : 𝒪[K])

instance : Fintype (IsLocalRing.ResidueField 𝒪[K]) :=
  Fintype.ofFinite (IsLocalRing.ResidueField 𝒪[K])
```

The instance `[ValuativeRel K]` is the class used to discuss the case where `K` is equipped with an equivalence class of valuations. `[IsNonarchimedeanLocalField K]` means that `K` is a nonarchimedean local field.

```
structure LubinTate.ℱ where
  toFun : PowerSeries 𝒪[K]
  trunc_two : toFun.trunc 2 = C π * X
  mod_pi : PowerSeries.C π ∣ toFun - PowerSeries.X ^ Fintype.card κ[K]
```

4.6 Height

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